

## SAMPOERNA UNIVERSITY – COURSE DESCRIPTION

### Course Catalog – General Education

Last updated: Spring 2024

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| DCOM0302                                                                                                                                                                                                                                                               | EAP Communication Skills IV                                          |
| Credit Hours                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                           | Fall (Offered for Intake 2023 and older)                             |
| Prerequisite                                                                                                                                                                                                                                                           | -                                                                    |
| Course Type                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| Designed to guide the students toward applying pronunciation, phrasing, and intonation of oral American English to communication situations in commercial, academic, and social settings. Involves interview presentation and emphasis on developing listening skills. |                                                                      |

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| DCOM0303                                                                                                                                                                                                                                                                                                                                        | EAP Reading IV                                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                    | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                    | Fall (Offered for Intake 2023 and older)                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                    | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| This is a high-intermediate ESL reading course designed for students in English for Academic Purposes (EAP) programs. It emphasizes vocabulary and comprehension at a high-intermediate level. Placement in this course is determined by successful or assessment tests and/or referral. Students must earn a "C" or higher to pass the course. |                                                                      |

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| DCOM0601                                                                                                                                                                                                                                                                                                                                       | EAP Grammar/Writing IV                                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                   | 6                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                   | Fall (Offered for Intake 2023 and older)                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                   | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| A high intermediate combined skills course for speakers of other languages designed principally to guide the students to the mastery of complex grammar and sentence structures, and basic paragraph writing. The requirement to move to the next level (EAP1540CU) is a "C" or higher. With a "D" or an "F", a student must repeat EAP0485CU. |                                                                      |

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|--------------|----------------------------------------------------------------------|
| GCOM1301     | EAP Composition I                                                    |
| Credit Hours | 3                                                                    |
| Term Offered | Fall (Offered for Intake 2023 and older)                             |
| Prerequisite | -                                                                    |
| Course Type  | Foundation   General Education Core   Critical Path   Upper Division |

A composition course in English for speakers of other languages. Designed principally to guide the student to the mastery of paragraph structure using various paragraph modes and the multi-paragraph essay. Grammar focuses on elements which closely tie in with composition, e.g. connectors and sentence combining. With a D or an F, a student must repeat EAP1540C.

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| GCOM1302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EAP Advance Reading                                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall (Offered for Intake 2023 and older)                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| This is an advanced reading course designed for English for Academic Purposes (EAP) students who have scored below 106 on PERT Reading or who require further development of reading skills in the English language. This course emphasizes literal, inferential, and critical comprehension skills, as well as analytical reading strategies and skills. In addition, the course emphasizes advanced vocabulary skills. Placement in EAP1520C is determined by successful completion of EAP0420C (a grade of C or higher) or placement test. Students must earn a C or higher to pass the course. |                                                                      |

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| GCOM1303                                                                                                                                                                                                                                                                                                                                            | EAP Composition II                                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                        | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                        | Fall (Offered for Intake 2023 and older)                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                        | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| This is an advanced composition course in English for speakers of other languages. Students compose essays in various rhetorical modes. Emphasis is placed on the writing process and the development of clear and logical ideas. Students apply advanced grammar and academic vocabulary skills. With a D or an F, a student must repeat EAP1640C. |                                                                      |

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| GCOM1304                                                                                                                                                                                                                                                                                                                                                                          | Composition I                                                        |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                      | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| Composition I is designed to learn and practice writing by creating original compositions, exploring basic rhetorical forms such as narration, exposition, and argumentation. Students will also develop research skills and learn to incorporate researched material through the writing process. This is a writing credit course that focuses on extensive writing and revision |                                                                      |

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| GCOM1305     | Composition II |
| Credit Hours | 3              |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Term Offered                                                                                                                                                                                                                                                                                                                                          | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                          | Composition I                                                        |
| Course Type                                                                                                                                                                                                                                                                                                                                           | Foundation   General Education Core   Critical Path   Upper Division |
| This course is designed to further develop a student's composition skills by building on the rhetorical modes/strategies learned in Composition I. The course requires students to observe the conventions of Standard American English and create documented essays, demonstrating the student's ability to think critically and write analytically. |                                                                      |

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| GCOM1306                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Introduction to Speech Communication                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Foundation   General Education Core   Critical Path   Upper Division |
| This course is designed to provide students with fundamental training and practical experience for researching, organizing, and delivering speeches in public situations. Topics include audience analysis, speech anxiety, critical listening, and preparation and delivery of informative, persuasive, and other possible types of public speeches in various cultural contexts with emphasis on academic and scholarly research. Students will also learn to effectively incorporate audio and visual aids/technologies for effective speeches. This is an international/intercultural competency course. |                                                                      |

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| GCOM1608                                                                                                                                                                                                                                                                                                                                         | EAP Advanced Writing                                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                     | 6                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                     | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                     | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| This Advanced Reading course is designed for English for Academic Purposes (EAP) students who require further development of writing skills in the English language. This course focuses on the whole system and structure of English language, paragraphs construction, and essay writing. Students must earn a C or higher to pass the course. |                                                                      |

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| GBUS2307                                                                                                                                                                                                                                                                                                                                 | Communications in the Workforce                                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                             | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                             | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| This course is designed to help students communicate more effectively in a general business setting. Students will practice analyzing, planning, managing, and executing both written and oral presentations. Special focus includes analysis, planning and all types of business documents to ensure appropriate content and structure. |                                                                      |

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| GHUM1207     | Kewarganegaraan |
| Credit Hours | 2               |

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| Term Offered                                                                                                                                                                                                                                                                                                   | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                   | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| This course is intended to qualify students with basic knowledge and skills as citizens who have national spirit and insight, have character, love the homeland, are democratic, obey the law, and could appreciate diversity, as the foundation for being able to contribute to the development of Indonesia. |                                                                      |

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| GHUM1208                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Bahasa Indonesia                                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| Bahasa Indonesia course is a general basic course that aimed to help the student in mastering Bahasa Indonesia effectively and productively. Students are expected to be able to understand and use Bahasa Indonesia on written or oral text in various context. Competencies that will be mastered by students are attitude and knowledge competency. In attitude competency, students are expected to show a positive attitude, actively contribute, responsibility, and be able to solve problems. In knowledge competency, students are expected to be able to master knowledge of Bahasa Indonesia and article writing. Thus, language skills that will be practiced in this course are reading, listening, writing, and speaking skills in relation to knowledge of Bahasa Indonesia and article writing. |                                                                      |

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| GHUM1301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Pancasila                                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| <p>In principle, Pancasila is the basis, view of life and ideology of the state, which is expected to unite the diversity of the nation in terms of ethnicities, religions, races, and languages. The birth of Pancasila was created through a long critical process politically, socially, and culturally. Pancasila consists of five core principles: (1) Ketuhanan yang Maha Esa/Belief in the one and only God, (2) Kemanusiaan yang Adil dan Beradab/ A just and civilized humanity, (3) Persatuan Indonesia/Unity of Indonesia, (4) Kerakyatan yang Dipimpin oleh Hikmat Kebjaksanaan dalam permusyawaratan/perwakilan/Democracy, led by the wisdom of the representatives of the people, (5) Keadilan Sosial bagi Seluruh Rakyat Indonesia/Social justice for all Indonesian people.</p> <p>However, the ideal goals of the Indonesian state and nation as stated in the Pancasila are not always in line with the reality on the ground. Conflicts between religions, poverty, discrimination, anti- democracy, injustice are still happening. As future leaders, every student at Sampoerna University is expected to not only understand the history and concept of Pancasila but also to be able to implement them in everyday life, both as individuals, citizens, and as God's creatures. This course will discuss the history and philosophy as well as the processes surrounding the formation of</p> |                                                                      |

Pancasila as the basis of the Republic of Indonesia. With this course, students will learn about the perception of the basic principles of Pancasila and how Pancasila is implemented in everyday life.

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| GHUM1303                                                                                                                                                                                                                                                                                            | World Religion                                                       |
| Credit Hours                                                                                                                                                                                                                                                                                        | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                        | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                        | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| This course is a descriptive examination of the world's most popular religions. GHUM1303 is a writing credit course. Students must earn a minimum grade of C to meet the requirements of the Gordon Rule for writing. World Religion meets the International/ Intercultural competency requirement. |                                                                      |

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| GHUM2304                                                                                                                                                                                                                                                                                                                                                                                                                                               | Introduction to Literature                                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| This course exposes students to the study of literature and a range of widely recognized authors and works. Students will examine and interpret a diverse and representative body of works from genres such as short stories, poetry, creative non-fiction, plays, and novels. Upon successful completion of this course, students will be able to demonstrate an understanding of fundamental concepts and ideas in each of the major literary forms. |                                                                      |

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| GSCI1103L                                                                                                                                                                                                                                                                                                          | General Chemistry I Lab                                              |
| Credit Hours                                                                                                                                                                                                                                                                                                       | 1                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                       | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| Laboratory experiment for General Chemistry I. Topics covered include chemical measurements, stoichiometry, atomic structure periodic table, chemical bonding, inorganic compounds, nomenclature, formula writing, gases, liquids, solids, solutions acid-base" chemistry and ionic reactions and redox reactions. |                                                                      |

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| GSCI1303                                                                                                                                                             | General Chemistry I                                                  |
| Credit Hours                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                         | Fall                                                                 |
| Prerequisite                                                                                                                                                         | -                                                                    |
| Course Type                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| This is the first course in a two-semester sequence, GSCI 1303 General Chemistry I and GSCI 2304 General Chemistry II. This sequence includes two laboratories: GSCI |                                                                      |

1103L to be taken concurrently with GSCI 1303 and GSCI 2104L to be taken with GSCI 2304. The course is for students who have already had high school chemistry. Topics covered include chemical measurements, stoichiometry, atomic structure periodic table, chemical bonding, inorganic compounds, nomenclature, formula writing, gases, liquids, solids, solutions acid-base chemistry and ionic reactions and redox reactions. To enroll in this course, it is strongly recommended that students have had previous chemistry at the high school or college level.

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| GSCI1105L                                                                                                                                                                                                                                                                                             | General Physics with Calculus I Lab                                  |
| Credit Hours                                                                                                                                                                                                                                                                                          | 1                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                          | Spring (Offered for 2023 Intake and older)                           |
| Prerequisite                                                                                                                                                                                                                                                                                          | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                           | Foundation   General Education Core   Critical Path   Upper Division |
| General Physics with Calculus I Laboratory is a laboratory which allows students to able to collect and analyze data in a variety of experiments covering topics covered in its companion course General Physics with Calculus I. Students will create experiment reports using analysis in calculus. |                                                                      |

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| GSCI1405                                                                                                                                                                                                                                                     | General Physics with Calculus I                                      |
| Credit Hours                                                                                                                                                                                                                                                 | 4                                                                    |
| Term Offered                                                                                                                                                                                                                                                 | Spring (Offered for Intake 2023 and older)                           |
| Prerequisite                                                                                                                                                                                                                                                 | Calculus & Analytical Geometry I (GMAT1505)                          |
| Course Type                                                                                                                                                                                                                                                  | Foundation   General Education Core   Critical Path   Upper Division |
| General Physics with Calculus I is the first course in a two-semester sequence outlining mechanics, properties of matter, heat, and sound. Algebra, trigonometry, geometry, and vector methods will be used in the quantitative description of these topics. |                                                                      |

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| GSCI2104L                                                                                                                                                                                                                                                                                       | General Chemistry II Lab                                             |
| Credit Hours                                                                                                                                                                                                                                                                                    | 1                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                    | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                    | General Chemistry I Lab (GSCI1105L)                                  |
| Course Type                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| The laboratory course is the continuation of General Chemistry Lab 1 which provides hands on activities that develop basic laboratory skills while reinforcing concepts learned in General Chemistry II. The General Chemistry II Lab is more complex in term of concept and Laboratory skills. |                                                                      |

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| GSCI2304     | General Chemistry II                                                 |
| Credit Hours | 3                                                                    |
| Term Offered | Spring                                                               |
| Prerequisite | General Chemistry I (GSCI1303)                                       |
| Course Type  | Foundation   General Education Core   Critical Path   Upper Division |

This is the second course in a two-semester sequence, GSCL 1303 and GSCL 2304. This sequence includes two laboratories: GSCL 1103L which should be taken Concurrently with GSCL 1303 and GSCL 2104L with GSCL 2304. This sequence is for students who have already had high school chemistry. Topics covered include: This sequence is for students who have already had high school chemistry. Topics covered include thermochemistry, thermodynamics, kinetics, acid-base chemistry, chemical equilibria, electrochemistry, nuclear chemistry, coordination compounds and an introduction to organic chemistry. To enroll, you must have completed CHM 1045 with a "C" or better. To enroll, it is strongly recommended that students have had previous Chemistry at the high school or college level.

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| GSCI2106L                                                                                                                                                                                                                                                                                               | General Physics with Calculus II Lab                                                           |
| Credit Hours                                                                                                                                                                                                                                                                                            | 1                                                                                              |
| Term Offered                                                                                                                                                                                                                                                                                            | Fall (Offered for Intake 2023 and older)                                                       |
| Prerequisite                                                                                                                                                                                                                                                                                            | General Physics with Calculus I Lab (GSCI1105L) and General Physics with Calculus I (GSCI1405) |
| Course Type                                                                                                                                                                                                                                                                                             | Foundation   General Education Core   Critical Path   Upper Division                           |
| General Physics with Calculus II Laboratory is a laboratory which allows students to able to collect and analyze data in a variety of experiments covering topics covered in its companion course General Physics with Calculus II. Students will create experiment reports using analysis in calculus. |                                                                                                |

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| GSCI2406                                                                                                                                                                                                                                                                                                                                                                                                                                                | General Physics with Calculus II                                                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4                                                                                              |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall (Offered for Intake 2023 and older)                                                       |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                            | General Physics with Calculus I Lab (GSCI1105L) and General Physics with Calculus I (GSCI1405) |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                             | Foundation   General Education Core   Critical Path   Upper Division                           |
| General Physics with Calculus II is part one of a comprehensive course in physics outlining electricity and magnetism, using analysis in calculus. The basic formulation of electricity and magnetism is discussed thoroughly, both in static situations and within the circuit. Their unification as an Electromagnetism phenomenon is described conceptually. Students are encouraged to put their understanding of Calculus I and II in this course. |                                                                                                |

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| GSCI1302                                                                                                                                                                                                                                                                          | Introduction to Environmental Science                                |
| Credit Hours                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                      | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| GSCI1302 is a study of the physical environment, and man's impact on natural systems. This course includes ecological systems, Florida and Indonesia environments and geology, pollution and environmental regulations, renewable and nonrenewable resources, and sustainability. |                                                                      |



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| GSCI1108L                                                                                                                                                                                                                                                                                       | General Biology Lab                                                  |
| Credit Hours                                                                                                                                                                                                                                                                                    | 1                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                    | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                    | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| GSCI1108L introduces students to perform the practical exercise and scientific methods as a tool to demonstrate basic concept presented in general biology course (cell structures and functions, principle of biochemistry, genetics, cell divisions, prokaryotes, eukaryotes, and ecosystem). |                                                                      |

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| GSCI1308                                                                                                                                                                                                                                                                                           | General Biology                                                      |
| Credit Hours                                                                                                                                                                                                                                                                                       | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                       | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| GSCI1308 introduces the concept and principles of biology which include the structure of cell and function, the principles of biochemical, energy transfer, genetics, mitosis & meiosis, evolution, organism (prokaryotes, protists, fungi, plants, and animals), biotechnology and the ecosystem. |                                                                      |

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| GSCI1307                                                                                                                                                                                                                                                                                                                   | Microbes & Society                                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                               | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                               | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                               | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                | Foundation   General Education Core   Critical Path   Upper Division |
| This course introduces that microbe play an important role in societies. It includes the role of microbes in the biosphere, agriculture, industry, biotechnology, and the relationship of microbes with humans, diseases and sanitation. It will also focus on the positive and negative effects of microbes in Indonesia. |                                                                      |

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| GSCI3311                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Ecology                                                              |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| GSCI3311 covers the basic concepts, and theoretical principles of ecology: the study of the interactions of organisms with each other and their environment. The topics for this course are nature of ecology, climate, the aquatic environment, the terrestrial environments, plant and animal adaptation to environment, history of life, properties of populations, interspecific competition, predation, parasitism and mutualism and community structure and dynamics. |                                                                      |



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| ENGR3303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Environmental Ethics                                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course introduces environmental ethics in Indonesia. Students will be able to investigate and seriously consider how and why we should live as morally responsible toward nature. Students will also examine critical theoretical approaches, key thinkers, and social movements in contention over competing for environmental values. The student will be able to explore philosophical responses to questions such as: What makes something natural? What value is there to non-human entities? What obligations do we have to each other regarding the environment? How should we respond to catastrophic environmental change?</p> |                                                                      |

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| GSCI1201                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Total Wellness                                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Total Wellness emphasizes the importance of knowledge, attitudes, and practices relating to personal wellness. It is a course designed to expose students to a broad range of issues and information relating to the various aspects of personal wellness including physical, social, emotional, intellectual, spiritual, and environmental wellness. This course integrates personal wellness and fitness in both a classroom and exercise environment. Evolving current topics such as nutrition, disease prevention, stress reduction, exercise prescription, and environmental responsibility are integrated to enable the student to understand the lifelong effects of healthy lifestyle choices.</p> |                                                                      |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| GSOC2303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | General Psychology                                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| <p>General Psychology studies theories and scientific principles related to human behavior and mental processes. Topics include the scientific method, neuroscience, learning, memory, and thinking, emotions, motivation, life span development, personality, and social psychology. General Psychology is a writing credit course. Students must earn a minimum grade of C to meet the requirements of the Gordon Rule for writing. General Psychology meets the International/Intercultural competency requirement. Learning Outcome Critical Thinking Global Awareness Information Literacy.</p> |                                                                      |

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| GSOC2304                                                                                                                                                                                                                                                                                                                                                                                                                 | Social Problems                                                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                             | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                             | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is an examination of the major social problems found in our changing social environment. More specifically, students will be introduced to a variety of topics which may include inequality, marginalization and oppression based on class, race, ethnicity, education, age; violence in society; the changing family; social problems related to gender and sexual behavior; global social problems.</p> |                                                                      |

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| GSOC1301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Principles of Macroeconomics                                         |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course serves provides the foundation of macroeconomics. This course will discuss theories and concepts of aggregate demand and aggregate supply, unemployment, inflation, aggregate measures of economics performance, fiscal policies, and monetary policies. The students are expected to understand the role of households, firms, and government in the economy. In addition, this course will help students to evaluate policy and business decisions in a logical and trained manner.</p> |                                                                      |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| GSOC2306                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | U.S History since 1877                                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This survey course of American history since 1877 provides students with a general history of the political, economic, cultural, and social development of American society. Special emphasis is placed upon U.S. expansion, progressivism, foreign relations, social movements, and political developments at the turn of the twentieth century and beyond. Students will also study the introductory concepts of history reading, writing, and methods. Through our efforts, the class will consider the importance of understanding American history and culture in a global context. For students taking this course for credit at Broward College, this is a writing credit course. Students must earn a minimum grade of C to meet their requirements of the Gordon Rule for writing.</p> |                                                                      |

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|--------------|----------------------------------------------------------------------|
| GMAT1301     | College Algebra                                                      |
| Credit Hours | 3                                                                    |
| Term Offered | Fall, Spring                                                         |
| Prerequisite | -                                                                    |
| Course Type  | Foundation   General Education Core   Critical Path   Upper Division |

College Algebra is a course in which students will learn the concepts and applications of topics such as solving, graphing, and applying linear, and quadratic equations, and inequalities; exponential and logarithmic properties; linear, quadratic, rational, absolute value, square root, cubic, and reciprocal functions; operations, compositions, and inverses of functions; and systems of equations and inequalities.

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| GMAT1512                                                                                                                                                                                                                                                                                                                                                                                                                                            | Extended College Algebra                                             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| Extended College Algebra is a course in which students will learn extensively the concepts and applications of topics such as solving, graphing, and applying linear, and quadratic equations, and inequalities; exponential and logarithmic properties; linear, quadratic, rational, absolute value, square root, cubic, and reciprocal functions; operations, compositions, and inverses of functions; and systems of equations and inequalities. |                                                                      |

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| GMAT1511                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Precalculus Algebra & Trigonometry                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| This course is designed to satisfy the dual requirements of Precalculus and Trigonometry, thus preparing the student for the study of calculus. In this course the student will study various function families (e.g. polynomial, exponential, logarithmic, trigonometric) from both analytic and graphical viewpoints, and will use them to model real-life situations. The student will be exposed to additional topics that will deepen their mathematical understanding, including systems, matrices and determinates, sequences and series, parametric equations, and polar coordinates and equations. A graphing calculator may be required. Recommendation of the Mathematics Department or at least a grade of "B" in the prerequisite course is required. |                                                                      |

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| GMAT1304                                                                                                                                                                                                                                                                                                                                    | Calculus for Business, Social & Sciences                             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| This is a general education course which includes the college-level skills of calculus such as: functions, graphs, limits, differentiation, integration, average and instantaneous rates of change, and other applications. Recommendation of the Mathematics Department or at least a grade of "C" in the prerequisite course is required. |                                                                      |

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| GMAT1505                                                                                                                                                                                                                                                                                           | Calculus & Analytical Geometry I                                     |
| Credit Hours                                                                                                                                                                                                                                                                                       | 5                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                       | Fall, Spring (Offered for Intake 2023 and older)                     |
| Prerequisite                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| This course is the first of a three-course sequence in calculus. Topics include Limits and their properties, Differentiation, Applications of Differentiation, and Integration. Some sections of this course may require technological tools, such as a graphing calculator, GeoGebra, and Desmos. |                                                                      |

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| GMAT2506                                                                                                                                                                                                                                                                                                                                                                                                         | Calculus & Analytical Geometry II                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                     | 5                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                     | Spring (Offered for Intake 2023 and older)                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                     | Calculus & Analytical Geometry I                                     |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| This is the second of a three-course sequence in calculus. Topics include techniques of integration, conics, polar coordinates, indeterminate forms, L'Hopital's Rule, proper integrals, infinite series, parametric equations, improper integrals, volume, arc length, surface area, work, and other applications of integration. Certain sections of this course may require the use of a graphing calculator. |                                                                      |

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| GMAT2507                                                                                                                                                                                                                                                                                                                  | Calculus & Analytical Geometry III                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                              | 5                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                              | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                              | Calculus & Analytical Geometry II (GMAT2506)                         |
| Course Type                                                                                                                                                                                                                                                                                                               | Foundation   General Education Core   Critical Path   Upper Division |
| This is the third of a three-course sequence in calculus. Topics include vectors in 3 space, 3 dimensional surfaces, multivariate functions, cylindrical and spherical coordinates, multiple integrals, partial derivatives, vector fields. Certain sections of this course may require the use of a graphing calculator. |                                                                      |

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| GMAT1310                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Statistics                                                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall, Spring                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | College Algebra (GMAT1301)                                           |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| GMAT1310 A first course in statistical methods including such topics as collecting, grouping, and presenting data; measures of central tendency, position, and variation; theoretical distributions; probability; test of hypotheses; estimation of parameters; and regression and correlation. Use of statistical computer software and/or a scientific calculator (capable of performing 2-variable statistics) will be required. Recommendation of the Mathematics Department or at least a grade of C in the prerequisite course is required. |                                                                      |

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| GMAT2308                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Ordinary Differential Equations                                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Calculus & Analytical Geometry II (GMAT2506)                         |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| <p>The Ordinary Differential Equations (ODE) course is an introductory course on differential equations which focuses on first-order differential equations and its modelling, and higher-order differential equations and its modelling. The course aims to equip the students with basic differential equations knowledge and skills that are needed in applied mathematics (mathematical modelling, dynamical systems), the physical sciences, engineering, and the social and biological sciences. By mastery ODE, students will be able to develop further knowledge and skill of mathematical modelling for real-life problems.</p> <p>Students will learn mainly about how the derivatives calculus appears in equations and is used to represent or to be a model of real-world phenomena.</p> <p>By mastery the ODE topics, students will be able to develop further knowledge and skill of mathematical modeling for real life problems.</p> |                                                                      |

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| GMAT2309                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Linear Algebra                                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Calculus & Analytical Geometry I (GMAT1511)                          |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is an introductory course to discover and explore linear algebra, the study of system of linear equations and their transformation properties. It is an important area of mathematics which provides the mathematical basis for modern presentations of geometry, mathematical modeling, and other branches of mathematics. Linear algebra has applications to explain fundamental principles and simplify calculations in engineering, computer science, mathematics, physics, biology, economics, and statistics. This course introduces students to some basic concepts of linear algebra and their applications.</p> |                                                                      |

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| MTED3314                                                                                                                                                                                                                                                                                                                                          | Discrete Mathematics                                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                      | Calculus & Analytical Geometry I (GMAT1511)                          |
| Course Type                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is an introductory course to discover and explore discrete mathematics, the study of mathematical structures (objects) which are discrete. It is an important area of mathematics which provides the mathematical basis for the understanding of computers and modern computation. The course has applications to almost every</p> |                                                                      |



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conceivable area of study such as computer science, data networking, chemistry, biology, linguistics, geography, business, and the internet. This course introduces students to the strategy and methods in proof, construct algorithm and apply various methods of counting principles in real life.

## Course Catalog – English Language Education Program

Last updated: Spring 2024

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| EDUC1301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation of Teaching and Learning                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| <p>As part of the curriculum of the teacher education program at Sampoerna University, the Foundations of Teaching and Learning course is a course that invites teacher candidates to move from a student perspective to a teacher perspective through the examination of teaching, learning, and schooling. Throughout this process, teacher candidates clarify, refine, and apply their personal theories of teaching and learning in classroom contexts. Teaching is a complex activity in which teachers apply knowledge from multiple domains, and from personal and professional experience, to develop curriculum, enact instruction, and assess learning. Learning to teach is a lifetime affair. Two very large and complicated questions guide the study and discussions during this course:</p> <ol style="list-style-type: none"> <li>1. What is Teaching?</li> <li>2. What does it mean to be a teacher?</li> </ol> <p>Throughout the course teacher candidates will also explore the following guiding questions, among many others: What is learning? Why should we learn and what to learn? What is the essence of teaching? What myths do we hold about teaching? How do we envision the role of the teacher? What does it mean to be a reflective educator? How might teachers create safe and engaging learning environments? How do teachers understand students? What are conceptions of good teachers and good teaching? How has history shaped teachers and teaching in Indonesia today? What social, cultural, and political ideas/ policies/ actions affect teaching and learning today? What should teachers teach? Who chooses what teachers teach? What choices do teachers have? What are our beliefs about school, teaching, learning and children? How do your lived educational experiences align with those beliefs? How do I identify and develop my educational philosophy? How might an educational philosophy impact teachers' thinking, judgment, and action? Why do people become teachers? Do I want to be a teacher? What kind of educator do I aspire to be? What does it mean to teach in a continued state of reform? This course will help teacher candidates learn how these questions have been answered and how course participants themselves would answer the questions.</p> |                                                                      |

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| EDUC1302                                                                                                                                                                                                                                                               | Educational Psychology                                               |
| Credit Hours                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                           | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                           | -                                                                    |
| Course Type                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course module aims to provide a descriptive and critical overview of the subject areas. Major theories, and approaches to the scientific study of behavior and mental processes in educational and psychology settings. The topics that will be covered in</p> |                                                                      |



the course are Human Development: Physical, Cognitive, Social and Emotional; Learning Theories; and Educational Planning and Practice.

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| ELTE1303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Planning Assessment & Evaluation                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Student teachers will study different approaches to learning, teaching, and assessment. It helps students to develop appropriate strategies which might be adopted in their own context. Students will also learn about various forms of assessment and develop one or two selected forms of assessment.</p> <p>Throughout the course, student teachers will discuss the social and cultural contexts of teaching and learn the importance of presenting substantial evidence of good practice. Teacher candidates will be given opportunities to understand the strengths and limitations of differing representations of learning, teaching and assessment and consider how, goals, subject content, teaching, learning, and assessment activities can be appropriately aligned, and student learning be advanced.</p> |                                                                      |

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| ELTE1326                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Writing Convention & Grammar Analysis                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This class examines the fundamentals of effective and eloquent writing. This course is designed particularly for humanities and teacher education majors who are interested in improving their own knowledge of grammar, reviewing the rules of traditional grammar, mastering the basics of structural grammar, and applying these principles to their own writing to improve its correctness, clarity, and effectiveness. In addition, the students will also practice the stages of the composing process (planning, drafting, and revising), learning to focus, organize, and develop their ideas, with attention to purpose and audience, and in the process make their writing more vivid and convincing.</p> |                                                                      |

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| ELTE2307                                                                                                                                                                                                                                                                                                                                                                                                                             | Teaching English as a Foreign Language (TEFL) I                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                         | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course aims to give some initial knowledge on Teaching English as a Foreign Language, putting the information they have obtained in English Language Methodology into some practices related to Teaching English as a Foreign Language. It provides some strategic planning, observation, and evaluation on how to teach English skills and to develop lesson plans to teach skills with emphasis on a particular skill.</p> |                                                                      |

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| EDUC2303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Educational Research Methodology I                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Educational Research Methods 1 (ERM1) course provides students with basic concepts and procedures of educational research. One of the important competencies for the new generation of teachers in Indonesia is to be able to conduct research and systematically reflect upon their own practices. This course provides the teacher candidates with the critical foundations for becoming a new generation of teachers.</p> <p>Students will have the opportunity to develop scientific thinking skills which are critical in academic and research methods. Students will also exercise academic writing that involves literature review. This course introduces students to key steps in conducting research and the important concepts in research including but not limited to qualitative and quantitative approach, sampling, validity and reliability, and experimental and non-experimental designs. It is expected that students can select and utilize quality research products and, in the future, they will be able to conduct one.</p> <p>By the end of this course, students will be able to know and demonstrate how to effectively read and use research papers for their own research interest. This course is also designed to build the students' readiness in proposing a research project. The end product of this course is the beginning part of a research project proposal, which consists of research background, problem statement, research questions, hypotheses, and brief plan for data collection and analysis (the detailed lessons about the latter will be presented in Educational Research Methods 2 course). In writing this proposal, students need to be able to write argumentative essays and engaging literatures. Hence, this product will demonstrate their ability to explore a research topic of their interests critically and creatively.</p> |                                                                      |

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| EDUC2305                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Social Foundation of Education                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| <p>The purpose of this course is to explore the relationships between education, social, political, economic, and cultural issues in Indonesia. Schools are ingrained in our larger society and contribute greatly to societal constructions and issues that often influence teaching, learning, and educational opportunities. In particular, we will explore issues of historical background, cultural diversity, social class, gender, exceptionality, language, geography, and religion.</p> <p>In a teacher education program, Social Foundations of Education will provide teacher candidates "the conceptual and practical tools to advocate for their own students and communities, and assist them in understanding how their work is influenced by social and structural forces, school and community contexts, and their own life histories and belief systems" (Tutwiler et al., 2013, p.108). Together with Educational Psychology,</p> |                                                                      |

Social Foundations of Education forms the standards of qualities of professional teachers. In their professional practice, teachers are frequently “called upon to exercise sensitive judgments amidst competing cultural and educational values and beliefs” (Council for Social Foundations of Education, 1996). This is why they need to be informed by studies in the ethical, philosophical, historical, and cultural foundations of education.

Rather than reducing education to a formula for best practice, courses in the Social Foundations of Education challenge students to think deeply about the relationships between education (formal and informal) and society(-ies) at large. Social Foundations encourages educators to use “critical judgment to question educational assumptions and arrangements and to identify contradictions and inconsistencies among social and educational values, policies, and practices” (Council for Social Foundations of Education, 2004).

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| ELTE2304                                                                                                                                                                                                                                                                                              | Introduction to the Study of Language                                |
| Credit Hours                                                                                                                                                                                                                                                                                          | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                          | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                          | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                           | Foundation   General Education Core   Critical Path   Upper Division |
| This course is to provide an overview of the nature of language and a chance to do linguistic analysis. It will help increase students’ awareness of major areas of linguistics, social significance, universality, diversity, and complexity. It will also help improve students’ analytical skills. |                                                                      |

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| ELTE2305                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Language Acquisition and Development                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| Language Acquisition Development (LAD) seeks to introduce students to how language is acquired. The central aim of this subject is to understand how children in general acquire their first language so effortlessly as the first language acquisition, and how young and adult people learn English as a second or foreign language. The acquisition of language structures will be detailed in the discussion of contemporary philosophies with selected topics related to language use and ideology. Although at times we will touch on issues in phonological and discourse acquisition, the majority of the lectures will be on syntactic and semantic issues within normal first language acquisition, with some attention to relevant material from second language acquisition and studies of linguistic deviance. Note: It is recommended that students have a working knowledge of introductory linguistics concepts, as we will not have time to cover these concepts in class. |                                                                      |

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|--------------|-------------------------------|
| ELTE2308     | Classroom Language Strategies |
| Credit Hours | 3                             |
| Term Offered | Fall                          |

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|--------------------------------------------------------------------------|----------------------------------------------------------------------|
| Prerequisite                                                             | -                                                                    |
| Course Type                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| Developing the skills to teach using English as a Medium of Instruction. |                                                                      |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ELTE2310                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Teaching English as a Foreign Language (TEFL) II                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| Previous TEFL 1 gave initial knowledge on Teaching English as a Foreign Language and its foundation: knowledge on strategic planning, observation, and evaluation on how to teach English skills and to develop lesson plans to teach those skills. In these TEFL 2, the course aims to give some knowledge on integrating intercultural communicative competence and multiliteracy and incorporating the use of low and high technology more for increasing media and digital literacy. |                                                                      |

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| ELTE3318                                                                                                                                                                                                                                                                                                                     | English for Young Learners                                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                 | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                 | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                 | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                  | Foundation   General Education Core   Critical Path   Upper Division |
| This course aims to equip student teachers with the knowledge and practical skills to teach English to primary age students. It offers the opportunity to explore the characteristic of young learners and methods on how create interesting, engaging activities which embed the contents of other subjects to the lessons. |                                                                      |

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| EDUC3307                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Social Emotional Learning                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| Social-emotional learning (SEL) is a course that underscores the significance of five fundamental skills aimed at fostering self-awareness, self-control, social development, and overall personal advancement. By the end of this course, teacher candidates are expected to gain a deeper understanding of the importance of SEL in their personal lives as an individual and educator. They are also expected to be better equipped to guide their students in understanding, acknowledging, and empathizing with their emotions. These acquired skills are subsequently applied to assist students in making constructive, responsible choices, establishing frameworks to reach their objectives, and fostering positive relationships with others. |                                                                      |

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|--------------|-------------------------------------------|
| ELTE 2323    | Literature for Younger and Older Children |
| Credit Hours | 3                                         |
| Term Offered | Spring                                    |

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| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course aims to provide students with an in-depth overview of various children's texts from both Indonesian and international contexts, as well as the central concepts and theories in the field of children's literature. The course will also look at the changing concepts of children and childhood and how these are linked to the production and reception of texts for children. Grounded in pedagogical approaches, the course aspires to afford students the opportunity to engage with diverse subjects in children's texts and to evaluate their potential in addressing social and cultural issues in wider communities. It invites students to take literary and critical stances and reflect on the possibilities for their future career in education.</p> |                                                                      |

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| EDUC2304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Educational Research Methodology II                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Educational Research Methodology I (EDUC2303)                        |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course provides an understanding of research methods, statistical analysis, needs assessments, and program evaluation. The significance of research to advance the field of education, to inform evidence-based practice, and the use of findings to effect program motivation will be covered. The course is designed to prepare students in the Faculty of Education to conduct high-quality research using various approaches; including, qualitative, quantitative, single-case design, mixed methods, action research, and outcome-based research.</p> |                                                                      |

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| EDUC3306                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Education Technology                                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Educational Technology (EDUC 3306) will help the teacher candidates to reflect on the role of various forms of technology that they can play in the teaching/learning process, and how they can engage these processes in their classroom. Through the subject, the teacher candidates will also become skilled in some of the many technological tools used in today's schools. In addition, they will be exposed to basic principles and theories in educational technology and will be assisted in determining appropriate applications of these theories and techniques in educational settings. Hands-on exercises and collaborative projects will be necessary to reach a high degree of competence.</p> |                                                                      |

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|--------------|----------------------------------------------------------------------|
| ELTE3317     | Curriculum and Material Development                                  |
| Credit Hours | 3                                                                    |
| Term Offered | Fall                                                                 |
| Prerequisite | -                                                                    |
| Course Type  | Foundation   General Education Core   Critical Path   Upper Division |

Basic knowledge of EFL curriculum, syllabus, lesson plan, and teaching materials development. Teaching material evaluation, selection, adaptation, and development of instructional materials that best suit the needs of a syllabus.

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ELTE3315                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Contemporary Approaches in ELT                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is dedicated to learning about current issues in ELT in the light of the latest trends in foreign language teaching and learning. Reviewing the contemporary issues in ELT can add values to language teachers and learners. Meanwhile, it will also lead us to a position where we are now, and where we are headed to prepare teacher candidates for future challenges. The Contemporary Issues in ELT course covers a variety of topics of English language teaching methods to Indonesian speakers, linguistics, language acquisition, the use of ICT in ELT to global Englishes, and examines those issues from a range of perspectives by drawing on the latest research findings.</p> <p>Particular attention will be given to pedagogical implications of the globally changing nature of the English language and ELT. The students are encouraged to develop their analytical, critical thinking and creative problem solving, to be engaged in independent and reflective learning.</p> <p>The course aims at exploring substantial, functional, and practical parts of those issues and developing multi-stand points and critical review of the issues.</p> |                                                                      |

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| EDUC3308                                                                                                                                                                                                                                                                                                                                                                                          | Individual Differences                                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                      | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| <p>On this course, students will learn about inclusive education, understanding individual differences, barriers, and challenges as the result of them. They will analyze the implementation of inclusive in school context. They will also elaborate on issues related to inclusive education such as curriculum adaptation, assessment, school and family collaboration, and also bullying.</p> |                                                                      |

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| ELTE3324                                                                                                                                                                                                                                                                                                                            | Literacy, Culture & the Development of Critical Readers              |
| Credit Hours                                                                                                                                                                                                                                                                                                                        | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                        | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                        | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Literacy, Culture, and Development of Critical Readers equip students with critical reading and thinking skills as parts of the academic skills needed in the era of change and uncertainty, in which cultures seem to have become globalized and 'homogenized', yet at the same time remain complex and incommensurate. The</p> |                                                                      |

abilities to search for information, determine its validity and accuracy, evaluate, and synthesize the information provided by the digital world in its abundance, as well as the ability to contextualize the information in its cultural specificity, play a strategic role across different sectors in society. The course requires students to carry out a set of different tasks ranging from using the available e-resources to collect information and make a critical comparison of data to formulating their analysis' results and communicating them to others. The tasks will have to be completed individually or through teamwork by means of collaborative learning and by making use of various e-learning media.

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ELTE3328                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Discourse Analysis & Sociolinguistic                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course introduces teacher candidates to the nature of spoken and written text and its use in social contexts. This course not only focuses on the linguistics symbol used but also on all the features that accompany language. It is designed to help teacher candidates understand the intricate relationship between linguistic and paralinguistic content of human behavior. This course introduces teacher candidates to the nature of spoken and written discourse and how they are used in social contexts. Teacher candidates will learn about some important sociolinguistics concepts that are useful in their line of work as language teachers. The sociolinguistic concepts include linguistic repertoire, bilingualism, code-switching, diglossia, and register. The teacher candidates will also learn pragmatic concepts such as speech acts, speech events, cooperative principles, and implicature which are linked to the structure of a conversation. As candidate teachers, they are expected to be familiar with lesson plans and classroom talks; thus, the final stage of this course introduced them to classroom discourse analysis which can enhance their understanding of classroom talks. In general, the course is directed to more practical activities that involve data collection (of a written and spoken discourse) and analysis which requires discipline, meticulousness, and critical thinking. Teacher candidates are expected to conduct research on a particular topic in discourse analysis, using elicited techniques or natural data collected during the semester.</p> |                                                                      |

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| EDUC3309                                                                                                                                                                                                                                             | Seminar                                                              |
| Credit Hours                                                                                                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                         | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                         | Educational Research Methodology II (EDUC2304)                       |
| Course Type                                                                                                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| <p>A capstone experience to identify feasible research topics, to review literature, and then to develop a research seminar paper: a critical literature study, a research project, classroom action research, or an applied evaluation project.</p> |                                                                      |

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| EDUC3315 | Advanced Educational Technology |
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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| <p>In this course, we'll be looking at a growing and compelling field, learning with and from games. This course is composed of six units of one to two weeks in length. In the first unit, we'll start by comparing educational games across time to better understand how history has shaped the present state of the field.</p> <p>We'll look at what practitioners have made and learn about their prototyping and development processes. We'll also look at case studies focused on the design and implementation of learning games, and we'll explore some frameworks for design.</p> <p>Throughout this course, you'll broaden your general game-design skill set by completing key assignments. For your final project, you'll also be developing your own game and accompanying curriculum with the freedom to choose how much to emphasize each.</p> |                                                                      |

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| ELTE3314                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Technology in Language Learning                                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Teaching English Language Learners through Technology explores the use of computers and technology as pedagogical tools to aid in the appropriate instruction of English language learners (ELLs) across all content areas. There are various technologies and software programs regularly used in the classroom for all students that can also specifically aid ELLs. Thus, teaching English Language Learners through Technology provides successful strategies for varying levels of access—whether teachers have one computer in their classroom, have multiple computers, can only occasionally use a computer lab, or through synchronous and asynchronous methods.</p> |                                                                      |

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| ELTE3325                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Literacy & Learning in Content Areas (Multiliteracies)               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Shifting from the literacy foundation 'read to learn' focus, the course aims to extend teachers' knowledge and skills to the processes, products, and values of literacy in various areas of the curriculum, with particular emphasis on comprehension, vocabulary, and disciplinary literacy. With a nod to the critical role of literature in one's literacy development, this course also requires teachers to engage with literature and have a direct experience to apply literature for literacy teaching (literature-based literacy teaching). For one semester, the course will cover the following major topics: Reading, writing and the connections, multimodality (which includes oral visual and</p> |                                                                      |

technological communications, and the content areas), and issues with grammar (phonic, spelling, and words study).

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| ELTE3313                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | English for Specific Purposes                                        |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| <p>English for Specific Purposes - ESP (ELTE 3313) provides an introduction and strategies for teachers to create instructional objectives, materials, and methods developed on the basis of learners' needs and potential of interests. This course prepares the pre-service teachers with specific skills and vocabulary needed to identify correctly and specifically the needs of their learners' own field in order to be able to communicate effectively in the target situation.</p> |                                                                      |

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| ELTE4322                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Advanced English Skills                                              |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Advanced English is a course that concerns improving and enhancing learners' abilities in English as a foreign language (EFL). One assumption upon which the course rests is that the students should bring to their roles as advanced EFL learners a strong competence of the full range of EFL communicative abilities - including listening, speaking, reading, and writing - resulting from intensive and extensive learning activities.</p> <p>The main purpose of this course is to equip students to operate as university-caliber writers and presenters. Focus will be on developing original academic arguments, using appropriate textual evidence to support the arguments, the ability to use counterarguments to temper and strengthen ideas, and developing a sophisticated and elegant prose style. Besides that, students will also practice how to present effectively using attractive and clear audio visuals to convince their audience; including handling questions and responding to those clearly and accurately.</p> |                                                                      |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| EDUC4311                                                                                                                                                                                                                                                                                                                                                                                                                                        | Micro Teaching                                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                    | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course focuses on teaching practices in a small group. Students are required to design the lesson plan and materials as preparation for their teaching performances. The teaching and learning process follows the stages of lesson study that emphasize on the teaching session, evaluation, and reflection. Each student needs to conduct the practice twice as the midterm (as a team) and final performance (as an individual).</p> |                                                                      |

Components to assess are presentations about lesson study, micro-teaching skills, and a portfolio that contains the teaching materials, lesson plans, and reflection.

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| EDUC4614                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Capstone Project                                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Seminar (EDUC3309)                                                   |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Capstone Project is a 6-credit course offered to final-year students. Over the course, students in the project work individually under a lecturer advisory to identify and address challenges in teaching and learning process, and to take the opportunity to improve the quality of education in Indonesia. Through the capstone project students can work closely with a lecturer in an area of interest exploring the project topic and methodology. Through this process and experience, students will develop their research and also increase their capacities and professional skills.</p> |                                                                      |

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| EDUC4312                                                                                                                                                                                                                                                 | Social Internship / Kuliah Kerja Nyata (KKN)                         |
| Credit Hours                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                             | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                             | -                                                                    |
| Course Type                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Social Internship Program (Kuliah Kerja Nyata, KKN) is one of mandatory courses for Faculty of Education students. The course will facilitate students to immerse in society by having a project to contribute to solution of real-life problems.</p> |                                                                      |

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| EDUC4313                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Internship / School Experience Program                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is intended to prepare and equip pre-service teachers to learn to develop their skills, knowledge, and dispositions to be effective and transformative educators to join SEP (School Experience Program) that happens every semester at different stages. Following a learning-to-teach continuum, the pre-service teachers start to develop their skills from simple tasks such as classroom and/or school observations, assisting mentor teachers in lesson preparation, instruction, and assessment, co-teaching with mentor teachers, to developing inquiry skills (reflection, research).</p> <p>School Experience Program (SEP) is a series of mandatory activities for all teachers candidates in Faculty of Education in Sampoerna University. This program aims to give them opportunities to experience real circumstances at schools, to develop knowledge, skills, and dispositions needed to be professional, transformative, high-qualified teachers. By doing tasks in every stage, the students would have to develop specific skill sets starting from observing teaching and learning in classroom and school environment, assisting mentor teachers in preparing lesson plan, assessment</p> |                                                                      |



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tools, and other teaching materials, reflecting upon teaching process, developing lesson plans and teaching materials, co-teaching learning sessions, with mentor/homeroom teachers, leading teaching and learning processes, to preparing and conducting data collection and data analysis.

## Course Catalog – Accounting Program

Last updated: Spring 2024

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| GBUS1302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Principles of Accounting I                                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| <p>GBUS1302/ ACG2001U course provides an introductory study of the fundamental principles of recording, summarizing, and reporting the financial activities of proprietorships and corporations. This course also prepares the accounting student in the theory and techniques of accounting as prescribed by the International Financial Reporting Standards (IFRS) and US Generally Accepted Accounting Principles (US GAAP) which are necessary for the advanced courses. Students will be introduced to financial statements and the accounting cycle for a service and merchandise business. Students will examine the procedures and systems to accumulate, analyze, measure, and record financial transactions. Students will use recorded financial information to prepare a Statement of Financial Position, Income Statement, Statement of cash flows, and Statement of shareholders' equity of a corporation. Students will be briefly introduced to Notes of Financial Statements.</p> |                                                                      |

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| GBUS1303                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Principles of Accounting II                                          |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Principles of Accounting I                                           |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Foundation   General Education Core   Critical Path   Upper Division |
| <p>As the continuation of Principles of Accounting I, this course covers more topics about financial accounting. They include current and long-term liabilities, partnerships, fixed and intangible assets, investments, statements of cash flow, partnerships, and simple financial statement analysis. The course also reviews the application of the International Financial Reporting Standard (IFRS) and U.S. Generally Accepted Accounting Principles (GAAP) in each topic.</p> |                                                                      |

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| ACCT1327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Personal Finance                                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Calculus for Business, Social & Life Sciences                        |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Foundation   General Education Core   Critical Path   Upper Division |
| <p>ACCT1327 course various areas of personal economic problems with which all individuals must contend. Students will learn about buying on credit, renting vs home ownership, using bank services, borrowing money, and investing savings; selecting from various types of insurance coverage; obtaining investment information, investing in bonds and stocks; Social Security; income taxes; retirement planning and annuities; medicare; and estate planning, wills, and trusts.</p> |                                                                      |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| GBUS2304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Managerial Accounting                                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Principles of Accounting II                                          |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course deals with planning, controlling, as well as costing products, services, and customers. It focuses on management accountant issues such as assisting managers to make better decisions through the concept of “different costs for different purposes”. By focusing on basic concepts, analyses, uses, and procedures, instead of procedures alone, it can be recognized that cost and management accounting as a managerial tool for business strategy and implementation. This course gives the accounting students the analytical and problem-solving skills such as excel to leverage available information technology. Topics covered include financial statement analysis, job order costing, the process cost system, cost behavior, cost-volume-profit analysis, budgeting, profit analysis, responsibility accounting, differential analysis, capital investment analysis, and decision-making under uncertainty.</p> |                                                                      |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ACCT2301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Intermediate Accounting I                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Principles of Accounting II (GBUS1303)                               |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Financial Accounting is undergoing a period of unprecedented change. Many countries have adopted or have time lies to converge with or adopt IFRS. The global reach of the IFRS is extensive. What is critically important is that the students of today (who will be the accountants of tomorrow) should develop strong competencies in the IFRS and be aware of the impending changes in specific reporting areas and the difference between IFRS and local accounting standards that remain. Designed to equip the student with the required techniques in preparing and interpreting financial statements. Emphasis is on conceptual frameworks, financial statements, current assets, inventory, property, plant, equipment, and intangible assets.</p> |                                                                      |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| ACCT2302                                                                                                                                                                                                                                                                                                                                                                                                                            | Business Law                                                         |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                        | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                        | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                        | Introduction to Business (GBUS1301)                                  |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                         | Foundation   General Education Core   Critical Path   Upper Division |
| <p>The course provides the student with foundational information about the Indonesian legal system, dispute resolution, and its impact on business. The primary content areas will include general principles of law, the relationship between business and law, the relationship between law and ethics, contracts, sales, torts, agency law, intellectual property, business law in the global context, and conflict of laws.</p> |                                                                      |

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| GBUS1305                                                                                                                                                                                                                                                                                                                                                                                                                    | Computer Applications                                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Desktop computing proficiency is increasingly important in today's business world. This course is designed to assist students in validating their knowledge and ability to use word-processing, spreadsheet, and presentation graphics applications. Through case studies, students will learn to develop comprehensive solutions to various types of problems. Integration between applications will be emphasized.</p> |                                                                      |

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| GBUS1105                                                                                                                                                                                                                                                                                                                                                                   | Computer Applications Lab                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                               | 1                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                               | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                               | -                                                                    |
| Course Type                                                                                                                                                                                                                                                                                                                                                                | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Desktop computing proficiency is increasingly important in today's business world. This course is designed to assist students in validating their knowledge and ability to use database application. Through case studies, students will learn to develop comprehensive solutions to various types of problems. Integration between applications will be emphasized</p> |                                                                      |

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| ACCT2328                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Corporate Finance                                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Personal Finance (ACCT1327) and Introduction to Business (GBUS1301)  |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course presents the fundamental ideas of corporate finance with an intuitive approach while placing a strong emphasis on computing abilities. This course also gives students the ability to solve problems and make decisions while also assisting them in developing an intuitive understanding of fundamental financial concepts. Students get a deeper knowledge of fundamental corporate finance principles through the use of an intuitive method, which also help students to build the critical thinking skills required to use financial tools in real-world decision-making scenarios.</p> <p>Topics covered include the financial manager, financial system, financial reports, time value of money, discounted cash flows, valuation, risk &amp; and return, bonds, stock, capital budgeting, cost of capital, working capital management, raising capital, dividends, business growth, options, and international financial management.</p> |                                                                      |



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| ACCT2314                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Individual and Corporate Tax I                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Principles of Accounting II (GBUS1303)                               |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This Individual and Corporate Tax I course covers The General Tax Provisions and Procedures Law and Individual Income Tax. After attending this course, students are expected to:</p> <ol style="list-style-type: none"> <li>1. Able to explain the basics of taxation and general provisions on taxation.</li> <li>2. Calculate Income Tax (PPh 21) in accordance with applicable regulations.</li> <li>3. Calculate other Withholding Taxes in accordance with applicable regulations.</li> </ol> |                                                                      |

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| ACCT2304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Intermediate Accounting II                                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Intermediate Accounting I (ACCT2301)                                 |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is the continuation of Intermediate Accounting I. Accounting is undergoing a period of unprecedented change. Many countries have adopted or have time lies to converge with or adopt IFRS. The global reach of the IFRS is extensive. What is critically important is that the students of today (who will be the accountants of tomorrow) should develop strong competencies in the IFRS and be aware of the impending changes in specific reporting areas and the difference between IFRS and local accounting standards that remain. Designed to equip the student with the required techniques in preparing and interpreting financial statements. Its focus is liabilities, equity, revenue recognition.</p> |                                                                      |

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| ACCT2330                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Fundamentals of Entrepreneurship                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Business Law (ACCT2302)                                              |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course provides an introduction to the field of entrepreneurship and explores the various stages and components of the entrepreneurial process. The resource offers a comprehensive and pragmatic sequential manual for effectively initiating and expanding an entrepreneurial endeavor, emphasizing the identification of opportunities and the assessment of their viability. The course encompasses various subjects such as identification and evaluation of opportunities, ideation, conducting feasibility studies, developing business plans, analyzing industry and competitors, formulating business models, establishing ethical and legal frameworks, evaluating the financial viability and strength of a venture, assembling a capable team, securing funding, implementing marketing strategies, and devising growth plans.</p> |                                                                      |

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| ACCT2329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Business and Professional Ethics                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Business Law (ACCT2302)                                              |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| <p>Students will learn how to handle ethical and sustainable business transactions and decisions in this course. Companies must manage their social capital due to market completeness, global warming, consumer activity, and major business crises. Humans can think and act generously and cooperatively, which creates capital. Social institutions that promote pro-social conduct and prevent exploitation are needed. Strong social capital companies prioritize customers, employees, and environmental welfare in their decision-making and have strong morals and long-term views. Textbooks, reports, journals, and case studies on food and farming enterprises will teach students how to define, measure, monetize, report, and manage social capital. They must protect social capital due to their huge influence as fundamental human needs providers. Unfortunately, this regularly happens. Social difficulties in modern agricultural value chains include labor norms and human rights violations, excessive working hours, low earnings, and uneven treatment. Socially “blind” commercial practices can cause food waste, malnutrition, and social insecurity. Conventional pesticide use is increasingly linked to food poisoning. Pesticides also cause endocrine-disrupting chemicals (EDC), which are prevalent and harmful to humans. EDCs are compounds in the environment (air, soil, or water), food, personal care, and manufactured items that disrupt the endocrine system.</p> |                                                                      |

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| ACCT3315                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Individual and Corporate Tax II                                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Individual Corporate Tax I (ACCT2314)                                |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This Individual and Corporate Tax II course covers the subject matters of Income Tax Law, Value Added Tax Law and other types of taxes. After attending the course, students are expected to be able to:</p> <ol style="list-style-type: none"> <li>1. comprehend the general concept of income taxes and value added tax</li> <li>2. explain the application of different types of income taxes</li> <li>3. explain the application of tax invoice and input tax</li> <li>4. prepare corporate annual income tax return and value added tax return</li> <li>5. understand other types of taxes such as property taxes (PBB), acquisition duty of right on land and building (BPHTB) and stamp duty (Bea Meterai).</li> </ol> |                                                                      |

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| ACCT3322                                                                                                                                                                  | Foundations of Accounting Research                                   |
| Credit Hours                                                                                                                                                              | 3                                                                    |
| Term Offered                                                                                                                                                              | Fall                                                                 |
| Prerequisite                                                                                                                                                              | Composition II (GCOM1305) and Statistics (GMAT1310)                  |
| Course Type                                                                                                                                                               | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course introduces students to academic research concerned with accounting. It aims to enhance students' knowledge of the research process and methodology and</p> |                                                                      |

enable them to be aware of the problems associated with research. The student is introduced to the role and process of research, data collection, concept of measurement, data analysis, and how to make a scientific research proposal and research report, including how run the data using software package (e.g., SPSS, or STATA), and how to write research results particularly in accounting and finance fields using secondary data.

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|--------------|----------------------------------------------------------------------|
| ACCT3307     | Auditing I                                                           |
| Credit Hours | 3                                                                    |
| Term Offered | Fall                                                                 |
| Prerequisite | -                                                                    |
| Course Type  | Foundation   General Education Core   Critical Path   Upper Division |

This course is an introduction to auditing and other assurance services. The primary emphasis on this course is on auditor's decision-making process in both an audit of financial statements and an audit of internal control over financial reporting that covers determining the nature and amount of evidence the auditor should accumulate after considering the unique circumstances of each engagement.

Phase I concerns client acceptance and continuance. Once a client is accepted (or the audit firm decides to continue to provide services to a client), the auditor needs to perform risk assessment procedures to thoroughly understand the client's business (or update prior knowledge in the case of a continuing client), its industry, its competition, and its management and governance processes (including internal controls) to determine the likelihood that financial accounts might be materially misstated (Phase II). In some audits, the auditor also obtains evidence about internal control operating effectiveness through testing those controls (Phase III). Much of what most people think of as auditing, the obtaining of substantive evidence about accounts, disclosures, and assertions occurs in Phase IV. The information gathered in Phase I through III greatly influences the amount of testing to be performed in phase IV. Finally, in Phase V, the auditor completes the audit and makes a decision about what type of audit report to issue.

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| ACCT3316     | Advanced Financial Accounting I                                      |
| Credit Hours | 3                                                                    |
| Term Offered | Fall                                                                 |
| Prerequisite | Intermediate Accounting II (ACCT2304)                                |
| Course Type  | Foundation   General Education Core   Critical Path   Upper Division |

This course is designed to provide students with a higher and more thorough understanding of some specific comprehensive problems in financial accounting, including Equity Method of Accounting for Investments, Preparation of Consolidated Financial Reports, and intercompany profit transactions between parents and subsidiaries.

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|--------------|----------------------------------------------------|
| ACCT3406     | Accounting Information System and Internal Control |
| Credit Hours | 4                                                  |
| Term Offered | Fall                                               |

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| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Computer Applications Lab (GBUS1105L)                                |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Foundation   General Education Core   Critical Path   Upper Division |
| Accounting Information Systems collect record, store, and process data to provide information for decision makers. They can be very simple paper-and-pencil-based manual systems, very complex using the latest in computers and information technology or something in between. The course introduces business process from an accounting cycles' perspective, emphasizing the nature and relevance of accounting internal controls and the relationship of accounting systems to the functional areas of accounting. This study will help students in deepen their knowledge of accounting as information provider useful for decision making and allowing them to acquire skills of information systems and technology. |                                                                      |

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| ACCT3331                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Introduction to Startup and Innovation                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fundamentals of Entrepreneurship (ACCT2330)                          |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Foundation   General Education Core   Critical Path   Upper Division |
| The course covers concepts of uncertainty and forecasting, the role of trial and error, and the scientific method in entrepreneurial planning. Students engage in a deep look at alternative methodologies, including the lean startup methodology as well as business plans. It offers tools that can dramatically alter the likelihood of success, directing your effort towards activities that help build actual businesses rather than useless idle dreaming or ineffective planning approaches. The course is an interactive class, using a significant amount of experimentation in the delivery method and a structure that applies first-hand experience in the methodologies presented. |                                                                      |

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| ACCT3332                                                                                                                                                                                                                                                                                                                                                                                                                   | Tools and Techniques in Project Management                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                               | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                               | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                               | Fundamentals of Entrepreneurship (ACCT2230)                          |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                | Foundation   General Education Core   Critical Path   Upper Division |
| This course offers an opportunity for participants to investigate the various components involved in the development and implementation of a comprehensive project plan. The curriculum will cover a range of subjects, such as establishing a project charter, identifying project requirements, delineating, establishing, and overseeing project boundaries, as well as assessing the validity of project deliverables. |                                                                      |

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| ACCT3311                                                                                                                                                                                                                                       | Financial Statement Analysis                                         |
| Credit Hours                                                                                                                                                                                                                                   | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                   | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                   | Corporate Finance (ACCT2328)                                         |
| Course Type                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| This course will prepare students with skills to understand information given by companies in their financial statements and to use it for making sound financial decisions. Financial statements provide information on a company's financial |                                                                      |

position, financial performances, and company's cash flows. The course will provide tools to use this information to determine a company's financial strengths and weaknesses.

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| ACCT3210                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Accounting theory                                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Intermediate Accounting I (ACCT2301)                                 |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Foundation   General Education Core   Critical Path   Upper Division |
| <p>The objective of this course is to provide the student with the opportunity to obtain a sound knowledge of normative, positive, and critical theories of accounting. The course begins by examining the nature of theories and alternative forms of logic. The conceptual framework and key contemporary and historical accounting issues are examined, highlighting the role of theory in understanding current accounting standards, accounting practice and the use of accounting information by the myriad stakeholders in reporting entities. Throughout the course examples of the relationship between theories of accounting and decisions facing real people (accountants and financial statement users) are highlighted.</p> |                                                                      |

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| ACCT3309                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Auditing II                                                          |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Auditing I (ACCT3307)                                                |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is a continuation of materials discussed in the pre-requisite subject which covers three out of five phases of the whole process of Audit.</p> <p>Phase I concerns client acceptance and continuance. Once accepted the auditor needs to perform risk assessment to thoroughly understand the client's business (or update prior knowledge in the case of a continuing client), its industry, its competition, and its management and governance process (including internal controls to determine the likelihood that financial accounts might be materially misstated (Phase II&amp;III). That briefly covers the first three phases of the audit. The information gathered in the first three stages greatly influences the following last two phases. Much of what people think of as auditing, obtaining substance evidence about accounts, disclosures and assertions occurs in phase IV. The actual application will be applied to the five cycles in the transaction cycles. Starting from the Revenue cycles; Cash, Marketable Securities and complex Financial Instruments; Auditing of Inventory, Goods and Services and Accounts Payable; the Acquisition and Payment Cycle; Auditing of Long-Lived Assets up to the Auditing of Debt, Obligations and Stockholders Equity transactions (Phase V).</p> <p>After completing these five cycles this subject will be closed by the discussion of the activities required in completing the quality audit and the Audit Reports on Financial Statements. During the course it is emphasized that professional judgment, sufficient appropriate evidence and quality decisions are critical to conducting a quality</p> |                                                                      |

financial statement audit. Besides, professional skepticism, which can be challenging to maintain is at the heart of auditor judgment.

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| ACCT3334                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Integrated Reporting                                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Intermediate Accounting I (ACCT2301)                                 |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| <p>ACCT3334 Integrated thinking and integrated reporting (IR) have long gone hand-in-hand. It provides inspiring examples of how this IR process has supported the design and execution of strategies underpinning value creation for both investors and other key stakeholders. This course will prepare students to understand key points of Multinational Companies' Integrated Reports according to the International Integrated Reporting Framework 2021 (the IIRF 2021) issued by the International Integrated Reporting Council. Students will be able to identify the content of the reports and to reconcile the differences between the report and the minimum content required by the IIRF 2021. The minimum content covers (1) Capitals: Financial, Manufactured, Intellectual, Human, Social &amp; Relationship, and Natural; (2) Principles: Purpose, Strategy, Risks &amp; Opportunities, Culture, Governance, and Performance.</p> |                                                                      |

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| ACCT3333                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EDP Audit                                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Accounting Information System and Internal Control (ACCT3406)        |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course will provide students with an introduction to IT auditing, emphasizing the concepts through exercises and case studies. Internal audit professionals will develop knowledge of basic IT audit concepts that can be used to facilitate integrated audit efforts within their organization. The course will also provide students the opportunity to perform an audit of IT applications supporting key business processes, coordinate the assessment of IT risk with the evaluation of IT general controls, and perform a risk assessment and evaluation of controls over end-user computing applications.</p> |                                                                      |

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| ACCT3335                                                                                                                                                                                                                                                                                                                                                                   | E-Commerce Technology in Business                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                               | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                               | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                               | Fundamentals of Entrepreneurship (ACCT2330)                          |
| Course Type                                                                                                                                                                                                                                                                                                                                                                | Foundation   General Education Core   Critical Path   Upper Division |
| <p>An overview of Internet technology, including its commercial applications and the social changes it has brought about. This course examines the technological aspects of the internet, as well as the collaborative and commercial endeavors, online communities, digital media dissemination, and social networking that occur within the context of the internet.</p> |                                                                      |



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| ACCT4324                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Global Corporate Governance                                          |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Business and Professional Ethics (ACCT2329)                          |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| <p>The recent financial crisis and major business scandals reminded the public the importance of corporate governance and business ethics in investor protection and well-being of the capital markets. The aim of the course is to provide an introduction of the important theories of corporate governance. The course will cover the topics on corporate board, ownership structure, CEO pay and turnover, institutional investor, regulatory reforms and corporate social responsibility. This Global Corporate Governance course covers the subject matter of Global Corporate Governance principles, pillars, framework and reform.</p> |                                                                      |

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| ACCT4336                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Professional Internship                                                                                                                                                                             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                                                                                                                                                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fall                                                                                                                                                                                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Pancasila (GHUM1301), World Religion (GHUM1303), Bahasa Indonesia, (GHUM1208) Principles of Macroeconomics (GSOC1301), Social Problems (GSOC2304), Introduction to Environmental Science (GSOC1302) |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division                                                                                                                                |
| <p>Students undertake a significant experiential learning opportunity whether in profit or non-profit organization. The internship is a learning strategy that allow students to apply their knowledge during the class meeting in links an applied work setting. Through direct observation, reflection and evaluation, students are expected to gain about work environment, business process, accounting application and also soft skills that may enhance their competence as future professional accountants. Students will produce a critical reflection on their internship experience demonstrating how they have addressed specific learning goals.</p> |                                                                                                                                                                                                     |

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| ACCT4338                                                                                                                                                                                                                                                                                                                                                                  | Scientific Writing in Accounting                                                                                                                                                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                              | 3                                                                                                                                                                                                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                              | Fall                                                                                                                                                                                                |
| Course Type                                                                                                                                                                                                                                                                                                                                                               | Foundation   General Education Core   Critical Path   Upper Division                                                                                                                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                              | Pancasila (GHUM1301), World Religion (GHUM1303), Bahasa Indonesia, (GHUM1208) Principles of Macroeconomics (GSOC1301), Social Problems (GSOC2304), Introduction to Environmental Science (GSOC1302) |
| <p>This course is an independent learning process in a research project accompanied by a supervisor. Students need to write down various information and data obtained from their research field with proper academic writing and methodology. Scientific research writing can be started from writing research proposals, research reports, proceedings and journals</p> |                                                                                                                                                                                                     |



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| ACCT4221                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | International Taxation                                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Individual Corporate Tax II (ACCT3315)                               |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course provides a clear introduction to international taxation and offers in-depth discussions on several important areas of the subject in a global context. Topics of discussions will not only cover the traditional issues that arise in international tax such as double tax problem, permanent establishment and improper use of treaties, but also taxation of digitalization of the economy which has been the central issue in international tax for the last five years.</p> |                                                                      |

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| ACCT3317                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Advanced Financial Accounting II                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Advanced Financial Accounting 1 (ACCT3316)                           |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course is designed to provide students with a higher and more thorough understanding of some specific comprehensive problems in financial accounting, including Equity Method of Accounting for Investments, Preparation of Consolidated Financial Reports, and intercompany profit transactions between parents and subsidiaries. This course is designed to provide students with a higher and more thorough understanding of some specific comprehensive problems in financial accounting. Students are also expected to understand the standards, whether it's the national standard Indonesian (PSAK) and International (IFRS) that relates to topics like employee benefit, share based compensation, hedge accounting, interim financial statement, segment reporting and other special topics.</p> |                                                                      |

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| ACCT3325                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Anti-Corruption Law                                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Business Law (ACCT2302)                                              |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course examines the nature, causes, and moral implications of corruption, as well as strategies that can be applied to combat corruption. It uses case studies, debates, and items from the news to examine how corruption can affect effective governance not only in the public sectors but also in the business entities as well. The students will be invited to look at examples of corrupt activity where it is not always the case that corrupt activities are illegal, or even that illegal activities are corrupt. Since the focus of the course is corruption in general, rather than corruption in any specific context, examples, and cases will be drawn from a range of roles and professions, in both the public and private spheres.</p> |                                                                      |

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|--------------|---------------------------------|
| ACCT3318     | Strategic Management Accounting |
| Credit Hours | 3                               |

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| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Managerial Accounting (GBUS2304)                                     |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course examines the function of the strategic management accountant in dynamic organizations that operate in a global business environment. In this position, the professional accountant collaborates with the organization's management team and contributes to strategy development and implementation, intending to generate customer and shareholder value and a solid competitive position. This course is designed to identify and apply advanced management accounting techniques for decision-making. It enables accountants to identify any deviation while monitoring all the cost, revenue, profit, and investment centers for continuous improvement. The course includes discussions on forecasting and budgeting techniques, standard costing &amp; variance analysis, project appraisal methods, cost analysis, modern costing techniques, balanced scorecards,</p> |                                                                      |

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| ACCT2305                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Management Control System                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Managerial Accounting (GBUS2304)                                     |
| Course Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Foundation   General Education Core   Critical Path   Upper Division |
| <p>This course will prepare students with skill to understand how firm's managers go about designing, implementing, and using planning and control systems to implement a firm's strategies. It focuses on strategy implementation. It gives students knowledge, insight, and analytical skills on how a corporate senior executive design and the ongoing management systems that are used to plan and control firm's performance. It explains management strategies, responsibility centers, transfer pricing, budgeting system, management compensation, and behavior when using these management tools. Students will learn this management control system by analyzing actual cases prepared by the Harvard Business School and the Tuck School of Business at Dartmouth. The cases are not necessarily intended to illustrate correct or incorrect answers. Rather, they are designed to allow students to gain practical experiences in management control problems and defend his or her opinion in the classes.</p> |                                                                      |

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| ACCT4223                                                                                                                                                                                                                                                                                                                          | Strategic Business Tax Planning            |
| Credit Hours                                                                                                                                                                                                                                                                                                                      | 2                                          |
| Term Offered                                                                                                                                                                                                                                                                                                                      | Spring                                     |
| Prerequisite                                                                                                                                                                                                                                                                                                                      | Individual and Corporate Tax II (ACCT3315) |
| <p>This course examines tax strategy and planning topics related to making important business decisions. The course addresses business formations, operations, terminations, reorganizations, acquisitions, and divisions. The course also covers tax issues related to multi-state and multi-national business transactions.</p> |                                            |

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| ACCT4314     | Public Sector Accounting |
| Credit Hours | 3                        |
| Term Offered | Fall                     |

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| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Intermediate Accounting I (ACCT4313) |
| <p>This course provides the students with an understanding of International Public Sector Accounting Standards (IPSAS) with a focus on their practical application and interpretation. The lectures will include numerous examples of their application by governments and public sector entities that are issuing IPSAS-compliant financial statements. By drawing on examples, practice highlights on specific accounting and reporting issues, including their implementation in Indonesia, students are expected to comprehend how to apply accrual-based IPSAS as well as the implementation of public sector accounting in Indonesia.</p> |                                      |

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| ACCT4340                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Corporate Risk Management                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring                                                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Accounting Information Systems and Internal Control (ACCT 3406) |
| <p>The course is designed for master students and represents an upper-level course. This course discusses risk management from the perspective of non-financial corporations. The course examines various types of risks (e.g. market risks, credit risks and operational risks) and risk management procedures in the context of the general framework of enterprise-wide risk management (ERM). The emphasis of the course is on theoretical approach of creating value with implementation of ERM rather than on the technical details of statistical measurement and pricing of derivatives. The course considers issues of risk measurement, risk aggregation, performance evaluation, capital allocation and strategic decision making.</p> |                                                                 |

## Course Catalog – Management Program

Last updated: Spring 2024

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| GBUS1301                                                                                                                                                                                                                                                                                                   | Introduction to Business |
| Credit Hours                                                                                                                                                                                                                                                                                               | 3                        |
| Term Offered                                                                                                                                                                                                                                                                                               | Fall                     |
| Prerequisite                                                                                                                                                                                                                                                                                               | -                        |
| <p>This course provides a basic study of business activity and how it relates to our economic society. Topics covered include how businesses are owned, organized, managed, and controlled. Course content emphasizes business vocabulary, areas of business specialization, and career opportunities.</p> |                          |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| BUSI1301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Basic Economic Issues |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                     |
| <p>This course serves as an introduction to Basic Microeconomic Theory and Basic Macroeconomic Theory. The Basic Microeconomic principles explain the action of individual market participants, including consumers, firms (producers) and workers. Basic Microeconomic models are powerful tools for both business and policy decisions. The Basic Macroeconomic section of the course studies the national economy as a whole. In this course, students will explore the causes of economic growth, how to assess the overall health of a national economy, and how fiscal and monetary policies are used to address unemployment, inflation, and economic growth.</p> |                       |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| BUSI1302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Entrepreneurship |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Spring           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -                |
| <p>This subject introduces entrepreneurship to entrepreneurial processes. It provides a thoughtful, practical step-by-step guide to the process of successfully launching and growing an entrepreneurship venture with a focus on opportunity recognition and feasibility analysis. Topics covered include recognizing opportunities and generating ideas, feasibility study, business plan, industry and competitor analysis, business model development, the preparation of ethical and legal foundation, assessing a venture's financial strength and viability, building the venture team, funding, marketing, and planning for growth.</p> |                  |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| BUSI2303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Analytical Methods for Business |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Spring                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Statistics (GMAT1310)           |
| <p>This course/subject is designed as a procedure or a method for the analysis of some problem, status or a fact. Analytical methods are usually time-limited and task-limited. They are used once to solve a specific issue. In practice there are used a lot of quite simple analytical techniques that managers and analysts use during normal work. Often without naming it somehow. For such techniques, it is often enough just a system of “paper-pencil,” or general office equipment. They are based primarily on the experience of the person who uses them. There are also a number of specialized analytical techniques, which are overwhelmingly based on some mathematical model or which require certain equipment or tools.</p> |                                 |

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| BUSI3304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Research Method for Business |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Fall                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -                            |
| <p>This course is aimed at developing students' understanding of the academic research methods plays in providing timely and objective information that drives decision-making in both the public and the private sector. Topics covered include the developing research proposal, research process, research designs, using secondary data, exploratory and qualitative research techniques, survey methods, measurement, questionnaire design, sample selection, sample size, population estimates and hypothesis tests, descriptive statistics and association tests, among others.</p> |                              |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| MGMT1301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Business Ethics & Sustainability |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                                |
| <p>This course will prepare students with the knowledge to deal with ethical and sustainable issues in business transactions and decision-making. Pressures from market completion, global warming, and consumer activism as well as major business scandals have forced companies to manage companies their social capital. It is a capital that arises from the human capacity to consider others and to think and act generously and cooperatively. It involves supportive social structures that encourage pro-social actions and discourage exploitative behavior. Companies with strong social capital will show strong integrity and moral values as well as long-term views in their decision-making paying attention to the welfare of their main stakeholders: customers, employees, and the environment.</p> <p>Students will learn how to define, measure, monetize, report, and manage a company's social capital through textbooks, reports, journals, and case studies with a focus on food- and farming-related businesses. Due to their massive influence as</p> |                                  |

providers of basic human needs, they need to be careful not to damage or destroy social capital.

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| MGMT2302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Digital Marketing |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                 |
| <p>Technology can open up completely new markets and to radically shake up existing ones. The mainstream adoption of digital technology – the Internet, the software applications that run on it, and the devices that allow people to connect both to the network and to each other whenever, wherever and however they want to – promises to dwarf all that has come before it. It heralds the single most disruptive development in the history of marketing. This subject provides comprehensive coverage of digital marketing, especially the social media for marketing communication, commerce and customer support; one-to-one communication to many different receiving devices; mobile computing; search engine optimization; consumer behavior insights based on offline and online data combination; inventory optimization through CRM-SCM integration; a single-minded focus on ROI and associated performance metrics and the explosion of social media are all on the cutting edge of digital marketing and they continue to develop as important strategies.</p> |                   |

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| MGMT3306                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Integrated Marketing Communication |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                  |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Spring                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                                  |
| <p>This subject presents an overview of essential inputs that a manager would consider when developing an integrated marketing communications plan. It introduces the various components of the marketing communications mix and summarizes the essential concepts dealing with consumer behavior, organizational behavior, and target marketing. It examines planning considerations for traditional media choices and new media choices. Because organizations look for synergy, the objective is to integrate related marketing and marketing communications practices with the media strategies. Students will also learn on how to examine the role of various research procedures for evaluating the effectiveness of marketing communications programs.</p> |                                    |

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| MGMT3310                                                                                                                                                                                                                                                                                                                                                                                                | Treasury Management |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                            | 3                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                            | Fall                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                            | -                   |
| <p>This subject provides a comprehensive coverage of contemporary international finance issues and problems, recognizing and capitalizing on the unique characteristics of global markets. The subject embeds real-world mini cases throughout to apply concepts to the types of situations managers of multinational firms face. Topics covered include the global financial environments, foreign</p> |                     |



exchange theory and markets, foreign exchange exposure, and financing the global firm, among others.

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| MGMT3311                                                                                                                                                                                                                                                                                                                                                                                          | Introduction to Finance |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                      | 3                       |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                      | Fall                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                      | -                       |
| This course will discuss various fundamentals and principles of finance. This course will cover topics in time value of money, financial statements and ratios, stocks and bonds, capital budgeting, financing mix, and international finance. Students are expected to understand the important roles of finance in decision making in the level of individual, small businesses, and corporate. |                         |

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| MGMT3308                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Business Model Innovation |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Fall                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -                         |
| Business Model Innovation (BMI) is open for all business students to help them to develop and innovate business models. Following this course will help students to understand the concept of the business model and its practical implementation. This course focuses on how people develop value propositions according to consumer expectations. With this, an organization can plan how it can generate profit and manage relevant business operations. Within this course students will get a mixture of conceptual exploration, real-world examples, and guest lectures. Students will also develop skills in written report and oral presentations. |                           |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| MGMT3312                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Introduction to Human Resources |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                               |
| Human Resource Management links people-related activities to business strategy. The course develops a critical understanding of the role and functions of the various human resource activities in an organization, providing students with a comprehensive review of key HRM concepts, techniques and issues. Topics include job analysis and design, recruitment and selection, evaluation, performance management, occupational health and safety, and the strategic contribution of HRM to organizational performance and evaluating HRM effectiveness. Working with contemporary case studies, students not only engage in collaborative and individual work processes but use communication and discourse characteristic of the HRM context and environment. |                                 |



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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| MGMT3313                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Real Estate Finance and Investment |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                  |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                                  |
| <p>This course will introduce students to concepts, theories, and practices in real estate. Students will be exposed to multiple aspects of real estate terminologies, including legal and regulatory, market valuation and appraisal, finance and investment in real estate. The course will start with some basic terminology and legal aspects of real estate. From there, the course will discuss the valuation of real estate by assessing its risk and return. This will be followed by financing and investing for homeowners and/or commercial real estate. Moreover, students are expected to use extensively Microsoft Excel during the course.</p> |                                    |

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| MGMT3314                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Intermediate Accounting for Business Administration |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall                                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                                   |
| <p>This course equips future managers, leaders, and entrepreneurs with intermediate accounting skills, i.e. the skill beyond constructing financial statements. This course will develop your skills to understand and use financial statements to assess the impact of your business decisions such as operation, investment and financing decisions. This course will use a simpler version of accounting technique of constructing financial statements and assessing the impact of managers, leaders, or entrepreneurs' decisions in businesses.</p> |                                                     |

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| MGMT3315                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Microeconomic Analysis for Business Decisions |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                             |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spring                                        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -                                             |
| <p>This course serves as an analysis into microeconomics. Microeconomic analysis explains the action of analysis in individual market participants such as supply analysis, demand analysis, and equilibrium analysis. Including analysis of consumers, and firms (producers). This course focuses upon a range microeconomic principles analysis, their use by economists in economic analysis and their relevance in the global economy. This course builds on principles economics by strengthening the students' knowledge of the fundamental theories related to micro level economics using advanced analytical tools and deeper level of discussions.</p> |                                               |

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|--------------|-----------------------------|
| MGMT3316     | Basic Operations Management |
| Credit Hours | 3                           |
| Term Offered | Spring                      |
| Prerequisite | -                           |

This subject provides a comprehensive coverage of the activities involved in operations management. The focus is on providing a fundamental working knowledge of the operations side of the firm. This subject presents a broad introduction to the field of operations in a realistic, practical manner. Operations management includes a blend of topics from accounting, industrial engineering, management, management science, and statistics. The subject first introduces the area of operations management including managing projects, forecasting demand, the global environment and operations strategy. The subject then focuses on designing operations beginning with the product design function, quality management and international standards as well as statistical process control. Additional topics include process design, location, layout and job design, managing the supply chain, and managing inventory.

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| MGMT3317                                                                                                                                                                                                                                                                                                                         | Integrating Business Fundamentals with Ethics and Law in Management |
| Credit Hours                                                                                                                                                                                                                                                                                                                     | 3                                                                   |
| Term Offered                                                                                                                                                                                                                                                                                                                     | Spring                                                              |
| Prerequisite                                                                                                                                                                                                                                                                                                                     | -                                                                   |
| The course provides the student with foundational information about the concept of ethics and its impact on business. The major content areas will include the general principle of ethics, differentiation between ethics and law, the relationship of business law, and ethics, and conflicting values surrounding a business. |                                                                     |

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| MGMT3318                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Economic Strategy for Business Decision |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                       |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                       |
| Economic strategy is a relatively new and rapidly developing area of economic consulting, involving the application of economic principles and methods to provide clients with unique insights aimed at addressing specific issues/problems and/or enhancing their long-term performance. Illustrative examples showing how clients may benefit from economic strategy consulting assignments include, but are not limited to, the following: Economic impact studies, Competitor analysis, Economics-based investment development, Change management, Assessment of the cross-border impact of public policy initiatives, Econometric analysis of advertising revenue, Formulation of economic development strategies, New market entry strategies, Implementation of economic development strategies, Competition and regulatory effects of new product development, State asset divestment strategies, Scoping studies, Bringing forward competition impact analysis of proposed transactions, Behavioral and experimental economics. |                                         |

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|--------------|-------------------------|
| MGMT3321     | Introduction to Startup |
| Credit Hours | 3                       |
| Term Offered | Spring                  |
| Prerequisite | -                       |

This course is an introduction to business startup development with main goal for students to understand the importance of business model, especially at the time when millennials and Gen Z prefer to start their own startup. This course is an introduction to many dynamic activities of the startup ecosystem. This course will allow students to gain a better understanding of what startups are all about, how a startup operates, and which business functions are needed in any startup venture.

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| MGMT3323                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Portfolio Theory and Analysis |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                             |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Fall                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                             |
| This course will introduce knowledge of capital markets and investment instruments with respect to theories, concepts, and analysis. This course focuses on the introduction of theories and concepts in financial investments, particularly on the analysis and valuation of stocks, bonds, and options. This course also covers portfolio theory (Markowitz Efficient Frontier) and study various methods (models) on pricing assets, such as Capital Asset Pricing Model (CAPM) and Arbitrage Pricing Theory (APT). |                               |

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| MGMT3324                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Credit & Lending Management |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall                        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                           |
| This course discusses the basic principles and management practices of banking. More specifically, the subject includes an overview of the banking business and banking management, the types of banking risk and capital management, assets management, general management, profitability management and liquidity management in banking. The discussion also includes ALMA (Asset - Liability Management) as well as off -balance sheet transactions, fee- based income and international banking. In addition, it also discusses topics related to the provisions of the current prudential and banking policy, as well as issues related to the stability of the financial system at the domestic and global levels. |                             |

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| MGMT4325                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Organization Behavior and Management |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                    |
| The overall aim of this subject is to ensure that human behavior contributes towards growth of the organization and greater efficiency is achieved. This subject investigates the impact that individuals, groups, and structure have on behavior within organizations to improve an organization's effectiveness. Workshops/Tutorials will emphasize the application of theories, through the use of exercises and case studies. Topics covered include diversity, attitudes and job satisfaction, emotions and moods, personality and values, perception, motivation, work teams, communication and leadership, power and politics, and conflict and negotiation. At a broader level, |                                      |



studying this subject will increase your awareness of issues such as: employee concerns during recessions, the benefits of cultural intelligence, happy workers are productive workers, ethical choice in workplace, organizational culture, East-West differences, gender issues in workplace, how motivation relates to job satisfaction, team culture, listening skills, ethical leadership, office politics, and emotions across cultures.

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| MGMT4326                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Introduction to Marketing |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                         |
| Introduction to Marketing is a foundation course required by all business majors. The primary goal of this course is to introduce the student to the key concepts that are important for the creation value of successful interfaces between the firm and its customer constituents. As an additional objective, this course will familiarize the student with terminology and common practices to professional marketing, especially when many marketing approach are changing in the disruption world. Upon the conclusion of this course, the student shall understand fundamental marketing issues especially the changing nature of Business and consumer, the place of marketing in the firm, and the different functions of marketing. |                           |

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| MGMT4327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Business Communication |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Fall                   |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -                      |
| This subject covers essential topics for effective business communication and exposes students to a wide variety of communications mediums used by today's businesses including innovative mediums such as blogging, podcasting, wikis and video as well as more traditional means of communication. The subject focuses on how to design and deliver oral and online presentations for internal and external audiences and addresses the opportunities and challenges of communicating in a diverse business world. Strong emphasis is placed on improving student's intercultural communication skills. |                        |

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| MGMT4328                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marketing Analytics |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Fall                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                   |
| This course is an introduction to marketing analytics. Today, companies heavily rely on data-driven marketing to better understand the needs of their customers. Through various data collection methods, they gather data on purchase behavior, social relationships, or attitudes. By analyzing such data, firms gain market insights and can enhance marketing decisions such as segmentation and targeting of customers, positioning of products based on customer preferences, or developing the right |                     |

marketing mix. However, marketers lack knowledge on how data can be collected and how basic research techniques can be applied to analyze it.

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| MGMT4329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Using and Managing Information Systems |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Spring                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                                      |
| <p>The pervasiveness of computer-based information technology (IT) in nearly all aspects of business means every individual in business needs to understand enough about IT to incorporate technology effectively into their enterprise. The content will introduce you to many concepts of information technology in business and organizations. Information systems are firmly rooted in an organizational context, so this is not merely a “technical” or “computer” course; this is a business course which involves the understanding of concepts and application of technical skills. Information systems are increasingly becoming more strategic and intertwined with the social systems in organizations, so this course will have a “strategic business” and “socio-technical” rather than a purely techno-centric perspective.</p> |                                        |

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| MGMT4330                                                                                                                                                                                                                                                                                                                                                                       | Macroeconomic and Global Institution Policy |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                   | 3                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                   | Spring                                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                   | -                                           |
| <p>This course discusses macroeconomics issues. Students are expected to understand the theories and concepts of aggregate demand and aggregate supply, unemployment, inflation, aggregate measures of economics performance, fiscal policies, and monetary policies. Moreover, students are able to discuss the role of households, firms, and government in the economy.</p> |                                             |

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| MGMT4331                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Innovation Principles |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -                     |
| <p>This course will teach you specific methods to assess and understand the industry, customers, and competitors for a new venture. You will then learn how to translate those insights into a winning venture idea, a business model, and set of distinctive new products and services. With this venture strategy in hand, you will then learn how to best raise venture financing, how to write a powerful business plan, and create a compelling pitch for investors. This course will show how these methods can be applied for corporate ventures as well as startups. For those students who wish to start their own venture, today or at some point in the future, this is the ideal course to learn real-world new venture planning. It will also be valuable for students who work for a large corporation and want to create entirely new businesses within that corporation.</p> |                       |



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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| MGMT4332                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Project Management |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                  |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                  |
| <p>The subject provides an overview of the management issues associated with selecting, implementing and terminating projects. This subject takes a holistic and integrated approach exploring both technical and managerial challenges to project management. The subject is divided into a series of major topic areas relating to different elements of project management. First, students are introduced to a strategic overview of project management, beginning with analyzing various components of project management such as project selection, planning, organizing and project control. The remaining topics introduce various theories to develop analytical and interpersonal skills useful to project managers. Particular emphasis is given in this subject to approximating the elements of project management and the challenges faced by project managers.</p> |                    |

## Course Catalog – Computer Science Program

Last updated: Spring 2024

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| ENGR1301                                                                                                                                                                                                                                                                                                          | Computer Programming for Engineering Applications |
| Credit Hours                                                                                                                                                                                                                                                                                                      | 3                                                 |
| Term Offered                                                                                                                                                                                                                                                                                                      | Spring                                            |
| Prerequisite                                                                                                                                                                                                                                                                                                      | -                                                 |
| This course provides an overview of the basics of programming and techniques used by computing professionals in a variety of application areas. Topics include computation, programs, algorithms, programming languages, and complexity, as well as how these concepts and techniques are used to solve problems. |                                                   |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| COMP2303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Digital Systems                                              |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Fall                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Computer Programming for Engineering Applications (ENGR1301) |
| This course is designed to introduce students to applied logic, which in turn introduces them to the basics of the electronics of digital systems. Students will learn how logic circuits are formed from the most basic components. They will also gain an understanding of how these basic components are combined to form larger systems. Topics in this unit include, but are not limited to: Number Systems, Logic Gates, Boolean Algebra, Arithmetic Operations, Logic Families, Counters, Registers, Microprocessor Basics, and an Introduction to Microcontrollers. |                                                              |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| COMP2424                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Data Structures & Algorithms                                 |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4                                                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fall                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Computer Programming for Engineering Applications (ENGR1301) |
| This course introduces a variety of data structures and their algorithmics. The course teaches students how to choose appropriate data structures and which type of algorithm is most likely to achieve a correct solution in the most efficient way. Topics will include arrays, lists, trees, hashing, sorting, stacks, heaps, minimum spanning tree, and shortest path algorithm. There will also be introduction to the fields of complexity theory, analysis of algorithms, and the notion of NP- completeness. |                                                              |

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| COMP2304                                                                                                                                                                                                                                                                                                                            | Computer Architecture and Organization |
| Credit Hours                                                                                                                                                                                                                                                                                                                        | 3                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                        | Spring                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                        | Digital Systems (COMP2303)             |
| This course introduces the students to computer architecture starting from top-level view of computer system and then taking closer view at the basic hardware components to construct more sophisticated circuits, which are then converted into memory units, processor units, as well as the whole computer systems. Topics will |                                        |



include overview of number systems and representation, computer arithmetic, the CPU, registers, bus architectures, instruction types, micro-operations, memory hierarchy, virtual memory, auxiliary memory, I/O peripheral devices and communication interfaces.

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| COMP1402                                                                                                                                                                                                                                                                                                                                                                                                                   | Object Oriented Programming             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                               | 4                                       |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                               | Spring                                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                               | Data Structures & Algorithms (COMP2424) |
| This course introduces students to the fundamentals of object-oriented programming languages through practical exposure to the Java programming language. Topics for this course will include an introduction to Java, fundamental programming structures, graphical user interfaces, and Unified Modeling Language. Emphasis will be placed on both the practical and theoretical aspects of object-oriented programming. |                                         |

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| GSCI2210                                                                                                                                        | Introduction to Optics and Thermodynamics                                                   |
| Credit Hours                                                                                                                                    | 2                                                                                           |
| Term Offered                                                                                                                                    | Spring                                                                                      |
| Prerequisite                                                                                                                                    | Calculus & Analytical Geometry II (GMAT2506) and General Physics with Calculus I (GSCI1405) |
| Introduction Optics and Thermodynamics is part one of a comprehensive course in physics outlining mechanical waves, thermodynamics, and optics. |                                                                                             |

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| COMP3423                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Mathematics for Engineering Applications |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4                                        |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Fall                                     |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Differential Equations (GMAT2308)        |
| This course is approximately one-half linear algebra and one-half probability and statistics. Linear algebra topics include matrix operations, systems of linear equations, determinants, Gauss-Jordan elimination, vector spaces, basis and dimension, projections, determinants, eigenvalues and eigenvectors. Probability and statistics topics include probability, random variables, density and distribution functions, sample mean and variance. An introduction to MATLAB and MATLAB projects. |                                          |

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| COMP3524                                                                                                                                                                                                                                                         | Basic Circuits                                                                               |
| Credit Hours                                                                                                                                                                                                                                                     | 5                                                                                            |
| Term Offered                                                                                                                                                                                                                                                     | Fall                                                                                         |
| Prerequisite                                                                                                                                                                                                                                                     | General Physics with Calculus II (GSCI2406) and Calculus & Analytical Geometry II (GMAT2506) |
| This course is designed to introduce students to basic circuits, Elementary, transient and sinusoidal analysis of linear circuits in the laboratory. Topics include passive sign convention, mesh and node analysis, Thevenin equivalents, op-amps, capacitance, |                                                                                              |

inductance, first and second order circuits, phasors, impedance, transformers, and PSpice simulation software.

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| COMP3408                                                                                                                                                                                                                                                                                                                                                                                  | Database Management Systems             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                              | 4                                       |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                              | Fall                                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                              | Data Structures & Algorithms (COMP2424) |
| This course introduces database management systems, relational databases and object-oriented databases. It covers relational algebra, SQL, store procedures, user-defined functions, cursors, an embedded SQL program, client-server interfaces, entity relationship diagrams, normalization, concurrency, transactions, database security, constraints, and object-relational databases. |                                         |

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| COMP2306                                                                                                                                                                                                                                                             | Operating Systems |
| Credit Hours                                                                                                                                                                                                                                                         | 3                 |
| Term Offered                                                                                                                                                                                                                                                         | Fall              |
| Prerequisite                                                                                                                                                                                                                                                         | -                 |
| This course introduces the students to a wider range of operating systems (OS) and their implementation details. Topics in this course include OS Architecture, OS Basics, OS Internals, Underneath the Framework, Resource Management, Security, and OS Case Study. |                   |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| COMP3313                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Web Programming (CS)                                                                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                                                                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                                                                                   |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Computer Programming for Engineering Applications (ENGR1301) and Database Management System (COMP3408) |
| This course is about designing and developing dynamic websites using PHP and MySQL. The course emphasizes hands-on coding exercises and critical thinking skills, with in-class practices and take-home assignments. Students will first go through the basics of web development and HTML. Then, the three tiers web architecture for web based application development will be introduced, discussed and implemented. Throughout the course students will learn how to build dynamic and interactive web applications by manipulating data, handling user input, and connecting to databases. |                                                                                                        |

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|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| COMP4322                                                                                                                     | Mobile Applications Development        |
| Credit Hours                                                                                                                 | 3                                      |
| Term Offered                                                                                                                 | Fall                                   |
| Prerequisite                                                                                                                 | Object Oriented Programming (COMP1402) |
| The course is designed to equip students with knowledge and skills to develop apps for mobile devices, particularly Android. |                                        |

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|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| COMP3309                                                                                                                    | Computer Networks & Data Communications |
| Credit Hours                                                                                                                | 3                                       |
| Term Offered                                                                                                                | Spring                                  |
| Prerequisite                                                                                                                | -                                       |
| The course is designed to equip students with knowledge and skills to understand data communications and computer networks. |                                         |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| COMP3326                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Circuit Theory                                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Spring                                                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Differential Equations (GMAT2308) and Basic Circuits (COMP3524) |
| <p>Electric circuit analysis using frequency domain techniques (Laplace Transforms, Fourier Series, and Fourier Transforms), sinusoidal, steady-state techniques to help with single-phase and three-phase power circuits, and time-domain techniques to help with convolution problems. These analysis techniques will be used to better understand circuit behavior, transformed networks, natural frequencies, poles and zeros, frequency response behavior (Bode plots), filters, and two-port networks.</p> <p>Circuit Theory is the second semester of electrical circuits in the Electrical and Computer Engineering curriculum. As such, it is critical that the topics, techniques, and skills in this course are learned at a level that allows you to successfully apply these skills and techniques in subsequent courses. You will become conversant in Laplace Transforms and their application to electric circuits.</p> |                                                                 |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| COMP3325                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Computer Architecture & Design                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Computer Architecture and Organization (COMP2304) |
| <p>This course focuses on the techniques of quantitative analysis and evaluation of modern computing systems and is appropriate for both advanced undergraduate and graduate students. We will build on introductory classes that show how a basic computer functions, and examine different techniques for improving computer performance, energy consumption, and usability. Emphasis will be placed on the major component subsystems of high-performance computers. We will discuss several topics, including pipelining, instruction-level parallelism, virtual memory, caches (memory hierarchies), multiprocessors, and advanced storage systems. Students will undertake a major computing system analysis and/or design project on topics of interest to the students. Hopefully, you will also have a lot of fun learning in this class.</p> |                                                   |

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| STEM4301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Internship for CS         |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Summer                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Minimum 100 credits taken |
| <p>Internships integrate knowledge and theoretical concepts learned from the academic setting with practice application and personal and career skill development in a professional setting. Students must first search for a company/ corporation/ institution/ establishment where they will conduct the internship. The business nature of the company/ corporation/ institution/ establishment must be related with computing disciplines.</p> <p>During the internship, students will perform the tasks given by their supervisor in the company/ corporation/ institution/ establishment and students will interact with other employees. The interaction between the students and the related people in the company/ corporation/ institution/ establishment where they perform their internship will form the personal and career skill development in a professional setting; while performing the tasks given by their supervisor will make the students apply the knowledge they gained in the academic setting.</p> |                           |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| STEM4302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Senior Capstone I                                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 3                                                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fall                                                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 110 credits hours already taken and Internship for CS (STEM4301) |
| <p>A culminating experience for Computer Science (CS) majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies.</p> <p>In this course students are introduced to do proper literature review to be able to identify and formulate a research problem for their final year project. Students will learn how to plan for doing a final year project by writing the capstone-1 report or the project proposal. Additionally, students need to be able to communicate their project effectively, both orally and in writing.</p> |                                                                  |

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| COMP3311                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Software Engineering                   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fall                                   |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Object Oriented Programming (COMP1402) |
| <p>This course gives a comprehensive analysis of software engineering techniques and its applications. Software engineering is a critically important technology for the future of mankind. Software engineers will need to develop this discipline so that they can create more complex software systems. Topics in this course include Introduction to Software Engineering; Software Process Models; Requirement Engineering; System Modelling and Prototyping; Design Engineering; Verification and validation; Software Project Management.</p> |                                        |

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| COMP4328                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Introduction to Communications |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                              |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Circuit Theory (COMP3326)      |
| <p>Analysis and design of analog and digital communication systems based on Fourier analysis. Topics include linear systems and filtering, power and energy spectral density, basic analog modulation techniques, quantization of analog signals, line coding, pulse shaping, AM and FM modulation, digital carrier modulation, and transmitter and receiver design concepts. Applications include AM and FM radio, television, digital communications, and frequency-division and time-division multiplexing.</p> |                                |

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| COMP4427                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Electronic Circuits       |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Circuit Theory (COMP3326) |
| <p>COMP4427 is a 4-unit course covering focus on the analysis and design of microelectronic circuits. You will learn about diodes and diode circuits, operational amplifiers, MOS and Bipolar transistor circuits, digital and analog circuits, small signal models and the use of PSPICE to simulate your circuit performance. During the lab you will learn how to breadboard microelectronic circuits and use electronic test modules for measurement and analysis.</p> <p>The course includes both lectures and lab. There is an emphasis on solving practical problems including the design aspects and measurements. I will also spend time during the lectures to cover prep work for the lab assignments since lab work will be a very important part of your grade and learning experience.</p> |                           |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| COMP4429                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Microprocessor Organization (with Lab)                                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 4                                                                                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                                                                              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Data Structures & Algorithms (COMP2424), Digital Systems (COMP2303) and Basic Circuits (COMP3524) |
| <p>This course is an introduction to microcontroller organization, hardware interfacing, and system design. Topics include but are not limited to: C Programming for Microcontrollers, Memory Organization and Addressing Modes, Interrupts, Timers, Parallel and Serial Interfacing, Analog-to-Digital Conversion, Overview of Common Peripheral Components, Event-driven Software Development, and Motor Control. In addition to lectures, students will have hands-on lab assignments that provide you with the opportunity to build and utilize the PIC24F platform. Students will also have a course project in which they will propose, design/implement, and present a self-selected project, subject to approval by the instructor.</p> |                                                                                                   |

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| STEM4304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Senior Capstone II           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Spring                       |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Senior Capstone I (STEM4302) |
| <p>A culminating experience for Computer Science (CS) majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies.</p> <p>Individuals or teams of students will use material taught in the CS curriculum to address a real-world client's needs or design and improve a system. Students will use a system design process, refine system requirements, proposed a methodology for problem solving, develop a plan and schedule, and present their work in oral presentation (oral defense) and Final Year Project exhibition. Students will communicate orally and in writing. A series of reports and presentations will monitor project goals, schedule, risk and progress. This course is a continuation of the Senior Capstone I.</p> |                              |

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| COMP3315                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Artificial Intelligence                                                                                                          |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                                                                                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Spring                                                                                                                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Data Structures & Algorithms (COMP2424), Discrete Mathematics (MTED3314) and Mathematics for Engineering Applications (COMP3423) |
| <p>In this course, students will be introduced to the basics of artificial intelligence. The course will cover the history, theory, and computational methods of artificial intelligence. Basic concepts include search, representation of knowledge and computational methods for reasoning. It will also cover intelligent agents; logical agents: propositional logic, first-order logic, logical inference; uncertainty; and learning. As part of AI techniques, machine learning will also be introduced in this course. Students will also gain experience of developing critical thinking and problem-solving skills in the context of intelligent machines.</p> |                                                                                                                                  |

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| COMP4330                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Knowledge System Engineering |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Spring                       |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                            |
| <p>Knowledge System Engineering is a course that teaches students how to make smart systems that can be used and process knowledge. The course goals include teaching different ways to show knowledge in a computer, and how to work with that knowledge. The course covers topics like reasoning, problem solving, dealing with uncertainty, making decisions, and learning new things. The course is for students who study Electrical and Computer Engineering, Computer Science, Industrial Engineering, or Management Information Systems. The course also shows different types of systems that can use knowledge, such as systems based on constraints, graphs, or logic. The course also teaches how to handle uncertain knowledge, use Bayesian networks, make decision trees, mine data, extract knowledge, and learn</p> |                              |

about neural networks and reinforcement learning if there is time. The course helps students learn how to make and use smart systems in different areas.

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| COMP4331                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Fundamentals of Information and Network Security                                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                                                                                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Data Structures & Algorithms (COMP2424) and Mathematics for Engineering Applications (COMP3423) |
| <p>This course adopts a principle of combining theory and practice. Students will learn the basic attacks and defenses by performing them via lab assignments. They will be introduced to a number of security tools to understand how they work and what security guarantee they provide. The experiments will be conducted in virtual machine environments. Students are expected to know basic C programming and Linux/Unix. This course covers topics: Introduction to security concepts and basic cryptographic building blocks. Implementation of fundamental security properties such as message and user authentication, confidentiality, privacy, anonymity, authorization, certification, non-repudiation, and revocation. Application of basic cryptographic primitives on building secure protocols and systems. Network security protocols.</p> |                                                                                                 |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| COMP4432                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Computer Aided Logic Design                                            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Data Structures & Algorithms (COMP2424) and Digital Systems (COMP2303) |
| <p>This course is an introduction to Computer-Aided Logic Design. This is a highly active research area, enabling the design of more complex digital systems. In this course we will focus on three areas: specification, optimization, and the use of software tools. The course will examine how to specify functionality at different abstractions, use industry-standard tools to synthesize and simulate hardware designs, and investigate some of the underlying optimization techniques utilized.</p> |                                                                        |



## Course Catalog – Industrial Engineering Program

Last updated: Spring 2024

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| GENG1301                                                                                                                                                                                                                                                                                                       | Introduction to Engineering |
| Credit Hours                                                                                                                                                                                                                                                                                                   | 3                           |
| Term Offered                                                                                                                                                                                                                                                                                                   | Fall                        |
| Prerequisite                                                                                                                                                                                                                                                                                                   | -                           |
| An introduction to the field of engineering. Students will learn about engineering careers, teamwork, ethics, engineering design, success in college, the teaching and learning process, and professional development. The class will also serve to prepare students for more advanced engineering coursework. |                             |

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| ENGR1301                                                                                                                                                                                                                                                                                                          | Computer Programming for Engineering Applications |
| Credit Hours                                                                                                                                                                                                                                                                                                      | 3                                                 |
| Term Offered                                                                                                                                                                                                                                                                                                      | Spring                                            |
| Prerequisite                                                                                                                                                                                                                                                                                                      | Calculus & Analytical Geometry I (GMAT1505)       |
| This course provides an overview of the basics of programming and techniques used by computing professionals in a variety of application areas. Topics include computation, programs, algorithms, programming languages, and complexity, as well as how these concepts and techniques are used to solve problems. |                                                   |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| IENG2302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Object Oriented Modeling & Design                           |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fall                                                        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Computer Programming for Engineering Application (ENGR1301) |
| This course introduces students to the modeling and design of complex systems using all views of the Unified Modeling Language (UML). The most effort will be in the problem domain (defining the problem). Some effort will be in the solution domain (producing hardware and/or software).<br>Students will understand how to develop models of systems, which may contain software and non-software components, and how to represent these models using the Unified Modeling Language (UML). Students will be able to design systems starting from the business and requirements model (via use cases) to the analysis model, to the design model, to the implementation model, and finally to the operational model. |                                                             |

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| IENG2301                                                                                                                                                                                                                                                                                                                      | Introduction to Systems & Industrial Engineering                                        |
| Credit Hours                                                                                                                                                                                                                                                                                                                  | 3                                                                                       |
| Term Offered                                                                                                                                                                                                                                                                                                                  | Fall                                                                                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                  | Introduction to Engineering (GENG1301) and Calculus & Analytical Geometry II (GMAT2506) |
| System modeling the elementary constructs and principles of system models including discrete-time, discrete-state system theory; finite state machines; modeling components, coupling, modes, and homomorphisms system design; requirements, lifecycle, performance measures and cost measures, tradeoffs, alternative design |                                                                                         |

concepts, testing plan, and documentation. Applications and case studies from engineering

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| IENG2304                                                                                                                                                                                                                                                                                                                              | Engineering Economy                                                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                          | 3                                                                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                          | Spring                                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                          | Introduction to Engineering (GENG1301) and Calculus & Analytical Geometry I (GMAT1505) |
| This course introduces students to the methods and modern techniques of engineering economy analysis for financial decision-making. Development of income, cash flow, and balance sheet statements are also covered. Topics include time value of money, valuation techniques, replacement analysis, and project acceptance criteria. |                                                                                        |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| IENG2305                                                                                                                                                                                                                  | Mathematical Foundations of Industrial Engineering                                                                                                        |
| Credit Hours                                                                                                                                                                                                              | 3                                                                                                                                                         |
| Term Offered                                                                                                                                                                                                              | Spring                                                                                                                                                    |
| Prerequisite                                                                                                                                                                                                              | Calculus & Analytical Geometry II (GMAT2506), General Physics with Calculus I (GSCI1405) and Computer Programming for Engineering Applications (ENGR1301) |
| Basics of mathematics and numerical computation in MATLAB, and their applications in solving engineering problems. (Pre-requisite: Calculus and Analytical Geometry II, General Physics I, Computer Program for Engineer) |                                                                                                                                                           |

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| IENG2103                                                                                                                                                                                                                                                                                                                                                                   | Industrial Engineering Colloquium |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                               | 1                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                               | Spring                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                               | -                                 |
| The course is designed to help students understand career opportunities for Industrial Engineering professionals. Students will interact with speakers and explore various roles of Industrial Engineering in solving real engineering problems. The course helps students select course options within the degree programs and helps focus on possible application areas. |                                   |

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| IENG3306                                                                                                                                                                                                                                                                                                                                       | Introduction to Engineering Probability and Statistics |
| Credit Hours                                                                                                                                                                                                                                                                                                                                   | 3                                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                   | Fall                                                   |
| Prerequisite                                                                                                                                                                                                                                                                                                                                   | Calculus & Analytical Geometry II (GMAT2506)           |
| Introduction to fundamental concepts and applications of probability and statistics in Industrial Engineering, with focus on how these concepts are used in experimental design and sampling, data analysis, risk and reliability analysis, and project design under uncertainty. Topics include basic probability concepts, random variables, |                                                        |

estimating parameters, regression, hypothesis testing, confidence intervals and Bayesian concepts.

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| IENG3309                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Manufacturing Processes for IE |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                              |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                              |
| Introduction to theory of manufacturing techniques and processes with an emphasis on metals, polymers and ceramics working processes for industrial applications. The course discusses various manufacturing processes and forms basic skills in the operation of standard machine tools and manufacturing processes. The instructional parts include introduction to materials engineering, casting, forming, conventional and non-conventional machining and joining. |                                |

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| IENG3109L                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Manufacturing Process for IE Lab |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                                |
| Introduction to theory of manufacturing techniques and processes with an emphasis on metals, polymers and ceramics working processes for industrial applications. The course discusses various manufacturing processes and forms basic skills in the operation of standard machine tools and manufacturing processes. The instructional parts include introduction to materials engineering, casting, forming, conventional and non-conventional machining and joining. |                                  |

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| IENG3307                                                                                                                                                                                                                                                                                                                                                                                    | Deterministic Operations Research                             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                | 3                                                             |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                | Fall                                                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                | Mathematical Foundations of Industrial Engineering (IENG2305) |
| This course introduces students to the theory, algorithms, and applications of optimization. The optimization methodologies include linear programming, network optimization, integer programming, and decision trees. Applications to logistics, manufacturing, transportation, and project management, including a team project in which students select and solve a problem in practice. |                                                               |

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| IENG3308                                                                                                                                                                                                                                                                                                                            | Software for Engineers                                       |
| Credit Hours                                                                                                                                                                                                                                                                                                                        | 3                                                            |
| Term Offered                                                                                                                                                                                                                                                                                                                        | Fall                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                        | Computer Programming for Engineering Applications (ENGR1301) |
| Rapid prototyping of decision support systems using VBA and Python with Excel and external packages to solve for optimization, build models and simulations, and create scheduling and forecasting tools. Decision support system examples include financial, supply chain, product portfolio and facility location and operations. |                                                              |

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| IENG3310                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Technical Writing         |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Composition II (GCOM1305) |
| <p>The course learns about preparing a technical report for engineering such as work reports, thesis, and presenting the report. This course is important because as an engineer, the students are not only able to do and solve many engineering problems but also communicate the result in the form of report. The course emphasizes the structure of a report, how to have critical thinking and step-by-step in preparing the report. In this course, students are trained to be hands-on in creating a report specific to their background. In addition, the use of Artificial Intelligence (AI) as a tool to create the report will also be studied.</p> |                           |

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| IENG3311                                                                                                                                                                                              | Probabilistic Models in Operations Research                       |
| Credit Hours                                                                                                                                                                                          | 3                                                                 |
| Term Offered                                                                                                                                                                                          | Spring                                                            |
| Prerequisite                                                                                                                                                                                          | Introduction to Engineering Probability and Statistics (IENG3306) |
| <p>The course covers fundamental probabilistic models and applications of operations research, which describe industrial systems and processes involving uncertain or random information or data.</p> |                                                                   |

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| IENG3315                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Quality Control and Six Sigma                                     |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Spring                                                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Introduction to Engineering Probability and Statistics (IENG3306) |
| <p>This Course is intended to give students background and a foundation in Quality Control and Six Sigma. The principles of Quality Control and Six Sigma and their applications in the design, development, manufacturing, delivery and service systems. We will discuss Quality Control and Six Sigma including: Definitions of Quality and Quality Improvement, Management Aspects of Quality Improvement, The DMAIC (Define, Measure, Analyze, Improve and Control) Process, Quality Function Deployment, Statistical Process Control, and Six Sigma.</p> |                                                                   |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| IENG3314                                                                                                                                                                             | Engineering Experimental Design                                   |
| Credit Hours                                                                                                                                                                         | 3                                                                 |
| Term Offered                                                                                                                                                                         | Spring                                                            |
| Prerequisite                                                                                                                                                                         | Introduction to Engineering Probability and Statistics (IENG3306) |
| <p>Design and analysis of experiments employing numerical and graphical methods. Topics include hypothesis testing, simple comparative tests, factorial designs, ANOVA analysis.</p> |                                                                   |

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| IENG3313                                                                                                                                                                                                                            | Production System Analysis                                                                                         |
| Credit Hours                                                                                                                                                                                                                        | 3                                                                                                                  |
| Term Offered                                                                                                                                                                                                                        | Spring                                                                                                             |
| Prerequisite                                                                                                                                                                                                                        | Introduction to Engineering Probability and Statistics (IENG3306) and Deterministic Operations Research (IENG3307) |
| This course introduces students to the production systems, quantitative methods for forecasting, aggregate planning, inventory control, material requirement planning, production scheduling, manpower planning and facility design |                                                                                                                    |

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| IENG3312                                                                                                                                                                | Integrated Manufacturing System                                                                                                                           |
| Credit Hours                                                                                                                                                            | 3                                                                                                                                                         |
| Term Offered                                                                                                                                                            | Spring                                                                                                                                                    |
| Prerequisite                                                                                                                                                            | Calculus & Analytical Geometry II (GMAT2506), General Physics with Calculus I (GSCI1405) and Computer Programming for Engineering Applications (ENGR1301) |
| Basic principles of basic primary and secondary manufacturing processes, integration of part design and manufacturing and integration of part design and manufacturing. |                                                                                                                                                           |

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| STEM4301                                                                                                                                                | Internship             |
| Credit Hours                                                                                                                                            | 3                      |
| Term Offered                                                                                                                                            | Semester 6 onwards     |
| Prerequisite                                                                                                                                            | Min. 100 credits taken |
| Specialized work on an individual basis, consisting of training and practice in actual service in a technical, business, or governmental establishment. |                        |

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| IENG4316                                                                                                                                                                        | Simulation Modeling and Analysis                                  |
| Credit Hours                                                                                                                                                                    | 3                                                                 |
| Term Offered                                                                                                                                                                    | Fall                                                              |
| Prerequisite                                                                                                                                                                    | Introduction to Engineering Probability and Statistics (IENG3306) |
| Discrete event simulation, model development, statistical design and analysis of simulation experiments, variance reduction, random variate generation, Monte Carlo simulation. |                                                                   |

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| IENG4317                                                                                                                                                                                                                                                                             | Human Factors & Ergonomics in Design                              |
| Credit Hours                                                                                                                                                                                                                                                                         | 3                                                                 |
| Term Offered                                                                                                                                                                                                                                                                         | Fall                                                              |
| Prerequisite                                                                                                                                                                                                                                                                         | Introduction to Engineering Probability and Statistics (IENG3306) |
| Human cognition, communication, information processing, and decision making. Work design, workload, situation awareness, and human error, Anthropometry, Work physiology, Workstation design, Human-Computer Interaction, Environmental Hazard, Standard, regulation, ethics in HFE. |                                                                   |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| IENG4318                                                                                                                                                                                                                                                                       | Project Management                                                |
| Credit Hours                                                                                                                                                                                                                                                                   | 3                                                                 |
| Term Offered                                                                                                                                                                                                                                                                   | Fall                                                              |
| Prerequisite                                                                                                                                                                                                                                                                   | Introduction to Engineering Probability and Statistics (IENG3306) |
| Organization strategy, Project Selection, Project definition and planning, Scheduling resources and cost, Managing project risk and procurement, Leadership and management of project teams, Progress and performance measurement, International Project, and Project Closure. |                                                                   |

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| IENG4319                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Human Side Organization |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3                       |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fall                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -                       |
| This course provides a comprehensive analysis of individual and group behavior in organizations. Its purpose is to provide an understanding of how organizations can be managed more effectively and at the same time enhance the quality of employees' work life. Topics include motivation, rewarding behavior, stress, individual and group behavior, conflict, power and politics, leadership, job design, organizational structure, decision making, communication, and organizational change and development. |                         |

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| STEM4302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Senior Capstone I                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Min. 110 credits taken and Internship (STEM4301) |
| A culminating experience for Industrial Engineering (IE) majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies. Individuals or teams of students will use material taught in the IE curriculum to address a real-world client's needs or design and improve a system. Students will use a system design process, discover system requirements, identify projects (problem and technical risks), and develop a plan and schedule. Students will communicate orally and in writing. A series of presentations will monitor project goals, schedule, risk, and progress. STEM4302 should be taken in the student's second to last semester. |                                                  |

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| IENG4320                                                                                                                                                                                                                                                                              | Supply Chain Management                                                                                          |
| Credit Hours                                                                                                                                                                                                                                                                          | 3                                                                                                                |
| Term Offered                                                                                                                                                                                                                                                                          | Spring                                                                                                           |
| Prerequisite                                                                                                                                                                                                                                                                          | Introduction to Engineering Probability & Statistics (IENG3306) and Deterministic Operations Research (IENG3307) |
| This course is intended to give students background and a foundation in the design and operation of supply chain systems. We will give an overview of supply-demand planning and operations and a detailed investigation into Supply Chain Distribution and Supply Chain Integration. |                                                                                                                  |

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| IENG4322                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Technical Sales and Marketing                                                                                    |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3                                                                                                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Spring                                                                                                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Introduction to Engineering Probability & Statistics (IENG3306) and Deterministic Operations Research (IENG3307) |
| <p>The intent of this course is to provide students with an understanding of the technical marketing and sales processes through a variety of experiences. Students will gain knowledge by attending lectures, participating in activities and projects, viewing videos, and reading course content material. Homework and classroom activities will broaden the students' understanding of the material. A major two-phase team project, as well as smaller activities and projects, will provide students with opportunities to simulate real-world technical sales situations.</p> |                                                                                                                  |

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| IENG4421                                                                                                                                                                                                                                                                                                                 | Embedded Computer Systems                                                                               |
| Credit Hours                                                                                                                                                                                                                                                                                                             | 4                                                                                                       |
| Term Offered                                                                                                                                                                                                                                                                                                             | Spring                                                                                                  |
| Prerequisite                                                                                                                                                                                                                                                                                                             | Introduction to Engineering (GENG1301) and Computer Programming for Engineering Applications (ENGR1301) |
| <p>Boolean algebra, combinational and sequential logic circuits, finite state machines, simple computer architecture, assembly language programming and real-time computer control. The computer is used as an example of systems engineering design; it is analyzed as a system, not as a collection of components.</p> |                                                                                                         |

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| STEM4303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Senior Capstone II              |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Spring                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Final Year Project I (STEM4302) |
| <p>A culminating experience for Industrial Engineering (IE) majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies. Individuals or teams of students will use material taught in the IE curriculum to address a real-world client's needs or design and improve a system. Students will use a system design process, refine system requirements, propose a methodology for problem solving, develop a plan and schedule, and present their work in oral presentation (oral defense) and Final Year Project exhibition. Students will communicate orally and in writing. A series of reports and presentation will monitor project goals, schedule, risk and progress. STEM4303 is a continuation of STEM4302 and should be taken in the student's last semester</p> |                                 |



## Course Catalog – Information System Program

Last updated: Spring 2024

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| INFS2301                                                                                                                                                                                                                                                                                                                            | Principles of Information Systems |
| Credit Hours                                                                                                                                                                                                                                                                                                                        | 3                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                        | Fall                              |
| Prerequisite                                                                                                                                                                                                                                                                                                                        | -                                 |
| As the organizations continue to operate in an increasingly competitive and global marketplace. This course provides a general idea to learn the use of information technology as the information systems are used in every profession. This course will deliver the content from the software, hardware, IoT, cloud computing etc. |                                   |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| INFS2303                                                                                                                                                                                                                                                                              | Enterprise Architecture |
| Credit Hours                                                                                                                                                                                                                                                                          | 3                       |
| Term Offered                                                                                                                                                                                                                                                                          | Spring                  |
| Prerequisite                                                                                                                                                                                                                                                                          | -                       |
| This course will explain the architecture, concepts, and methods that exist in the design and implementation of IT in enterprise scale companies. This course will help provide a specific understanding in requirements analysis, design, and implementation at an enterprise scale. |                         |

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| INFS2318                                                                                                                                                                                                                                                                                                                                                                     | Technopreneurship |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                 | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                 | Spring            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                 | -                 |
| This course will introduce students to the processes of motivation and innovation and of technopreneurship. This course will give an understanding of the principles of technopreneurship including developing business plan; marketing in the global information age; financing and marketing; innovation and creativity, financial management; and product identification. |                   |

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| INFS2330                                                                                                                                                                                                                                                                                                                                               | Economics of Information Systems |
| Credit Hours                                                                                                                                                                                                                                                                                                                                           | 3                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                           | Spring                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                           | -                                |
| This course introduces students to the methods and modern techniques of economic analysis for financial decision-making. Development of income, cash flow, and balance sheet statements are also covered. Topics include time value of money, valuation techniques, systems replacement analysis, and information systems project acceptance criteria. |                                  |

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| INFS2329                                                                                                                                                                                                                                                                                                                                                                                                | Fundamentals of Database Management System (DBMS) |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                            | Spring                                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                            | Data Structures & Algorithms (COMP2424)           |
| <p>This course introduces database management systems, relational databases, and object-oriented databases. The topics cover relational algebra, SQL, store procedures, user-defined functions, cursors, an embedded SQL Program, client-server interfaces, entity relationship diagram, normalization, concurrency, transactions, database security, constraints, and object-relational databases.</p> |                                                   |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| INFS3334                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cyber Ethics |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -            |
| <p>This course will introduce ethical and legal issues that arise in the field of information technology. New technology creates ethical challenges for individuals around the globe and applies to most people regardless of whether they are employed in the information technology field or a more traditional occupation. The study of Cyber Ethics provides a framework for making ethical decisions that professionals are likely to encounter in the workplace. The course will not only address ethics, but the legal, economic, social, cultural, and global impact of decisions made in the professional world.</p> |              |

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| INFS3332                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | IT Project Management |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                     |
| <p>The purpose of this course is to introduce selected topics, issues, problems, and techniques in information systems project management. Project management is implemented in most areas of business today; it is usually an assignment that provides essential exposure to decision-makers within a company. This course examines project management roles and environments, the project life cycle, and various techniques of work planning, control, and evaluation for project success. Upon completion of the course, the students should be familiar with the tools and processes of project management (the how-to of a project), as well as typical issues and problems encountered during real- world implementation in information system projects. After course completion the student should be able to address the issues that are important in effectively managing projects.</p> |                       |

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| INFS3331                                                                                                                                                                                                                                                                                                                           | Management Information Systems                               |
| Credit Hours                                                                                                                                                                                                                                                                                                                       | 3                                                            |
| Term Offered                                                                                                                                                                                                                                                                                                                       | Fall                                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                       | Fundamentals of Database Management System (DBMS) (INFS2329) |
| <p>This course introduces students to the development of information systems within the scope of the organization, starting with the concept of information systems, organization, the role of information systems in organizations, types of information systems, and the development and maintenance of information systems.</p> |                                                              |

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| INFS3333                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Introduction to Methods of Network Analysis |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                                        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Digital Systems (COMP2303)                  |
| <p>INFS3333 provides students with a methodology for analyzing networks by examining the network at its infrastructure, network and applications layers; exploring how they transfer data; investigating how network protocols work to enable communication; and probing and analyzing how the lower-level network flayers support the upper ones. Students will use hands-on labs and exercises to investigate and analyze network fundamentals. INFS333 meets the National Security Agency (NSA) Center of Academic Excellence in Cyber Operations (CAE-CO) academic requirements for networking principles.</p> |                                             |

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| INFS3304                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Information Systems Audit |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                         |
| <p>The aim of the course is to teach students to use their generalized knowledge in various fields of IT for estimation of information system (IS) characteristics and IS-related controls. This course is based on the program of CISA (Certified Information Systems Auditor) exam and introduces students to the concepts and practical methods of full cycle of IS auditing. This is the main feature of the course that distinguishes it from other similar disciplines studied in Russian and foreign Universities and aimed usually to separate categories of audit (e.g., audit of Information Security Management, IT strategy audit, Network Infrastructure audit etc.).</p> |                           |

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| COMP3314                                                                                                                                                                                                                                                                                                                                                  | Principles of Information Security |
| Credit Hours                                                                                                                                                                                                                                                                                                                                              | 3                                  |
| Term Offered                                                                                                                                                                                                                                                                                                                                              | Spring                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                              | -                                  |
| The course will introduce principal concepts in information security. Topics include Information Security Introduction. Information Security essentials, Legal, ethical, and professional issues. Information Security plan. Risk management. Information Security technologies. Cryptography. Physical security. Implementation. Personnel. Maintenance. |                                    |

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| INFS3307                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | E-commerce                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Spring                                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Management Information Systems (INFS3331) |
| This course provides the learner with an overview of the basic principles of electronic commerce and the related concepts, which are reflected in the current environment of the global economy. In the course, the student will develop a deeper understanding of the critical attributes of a successful participant in today's ever-changing markets. During this course the learner will utilize fundamental concepts learned in economic and marketing classes, integrated with computer skills to create an online business. Included in the content of this course are current issues related to electronic commerce issues. Security, digital money, and the evolving internet will be among some of these issues. Ethical, legal, and environmental issues will also be explored. |                                           |

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| INFS3336                                                                                                                                                                                                                                                                                                                      | Computational Thinking & Doing        |
| Credit Hours                                                                                                                                                                                                                                                                                                                  | 3                                     |
| Term Offered                                                                                                                                                                                                                                                                                                                  | Spring                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                  | Data Structure & Algorithm (COMP2424) |
| This course covers the basics of programming and techniques used by computing professionals in a variety of application areas. Topics include computation, programs, algorithms, programming languages, complexity, and computability. The course also explores how these concepts and techniques are applied in Informatics. |                                       |

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| INFS3335                                                                                                                                                                                                                                                                                                          | Introduction to Applied Computing |
| Credit Hours                                                                                                                                                                                                                                                                                                      | 3                                 |
| Term Offered                                                                                                                                                                                                                                                                                                      | Spring                            |
| Prerequisite                                                                                                                                                                                                                                                                                                      | Digital Systems (COMP2303)        |
| This course introduces students to computing concepts in hardware, software, networking, data processing, and other emerging technologies. Topics cover information representation, relational databases, system design, web development, and cutting-edge technologies for CPU, operating systems, and networks. |                                   |

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| STEM4301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Internship for IS |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Summer            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -                 |
| <p>Internships integrate knowledge and theoretical concepts learned from the academic setting with practice application and personal and career skill development in a professional setting. Students must first search for a company/ corporation/ institution/ establishment where they will conduct the internship. The business nature of the company/ corporation/ institution/ establishment must be related to computing disciplines. During the internship, students will perform the tasks given by their supervisor in the company/ corporation/ institution/ establishment and students will interact with other employees. The interaction between the students and the related people in the company/ corporation/ institution/ establishment where they perform their internship will form the personal and career skill development in a professional setting; while performing the tasks given by their supervisor will make the students apply the knowledge they gained in the academic setting.</p> |                   |

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| STEM4302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Senior Capstone I |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Fall              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                 |
| <p>A culminating experience for Information System (IS) majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies. Individuals or teams of students will use material taught in the IS curriculum to address a real-world client's needs or design and improve a system. Students will use a system design process, discover system requirements, identify projects (problem and technical risks), and develop a plan and schedule. Students will communicate orally and in writing. A series of presentations will monitor project goals, schedule, risk, and progress. Senior Capstone I should be taken in the student's second to last semester.</p> |                   |

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| INFS4305                                                                                                                                                                                                                                                                                                             | System Integration |
| Credit Hours                                                                                                                                                                                                                                                                                                         | 3                  |
| Term Offered                                                                                                                                                                                                                                                                                                         | Fall               |
| Prerequisite                                                                                                                                                                                                                                                                                                         | -                  |
| <p>This course discusses the design of a system by considering several aspects, evaluating a design result from several kinds of aspects, comparing several design results, determining the best design of all of them, and integrating the selected design in the form of a uniform technological architecture.</p> |                    |

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| INFS4337                                                                                                                                                                                                                                                                                                                                               | Data Analysis & Visualization |
| Credit Hours                                                                                                                                                                                                                                                                                                                                           | 3                             |
| Term Offered                                                                                                                                                                                                                                                                                                                                           | Fall                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                                           | -                             |
| <p>This course will learn about the basics of data science project based. Data science is emerged as the magnificent topic. frameworks, opportunities, and challenges of all activities related to Data. This course covers the basic principle of data science which including data classification, data clustering, regression, and forecasting.</p> |                               |

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| INFS4338                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cloud Computing |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fall            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -               |
| <p>This course provides students with an introduction to the concepts of cloud computing and a clear understanding of the variety, complexity, and capabilities of modern cloud platforms. The student will learn how to deploy web services in a cloud computing environment and how to scale capabilities using orchestration and automation technologies. The student will also be exposed to some of the fundamental security issues and challenges of a cloud computing environment and best practice approaches to mitigate those issues.</p> |                 |

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| INFS4342                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Senior Capstone   |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Senior Capstone I |
| <p>A culminating experience for Information System (IS) majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies. Individuals or teams of students will use material taught in the IS curriculum to address a real-world client's needs or design and improve a system. Students will use a system design process, refine system requirements, propose a methodology for problem solving, develop a plan and schedule, and present their work in oral presentation (oral defense) and Final Year Project exhibition. Students will communicate orally and in writing. A series of reports and presentation will monitor project goals, schedule, risk and progress. This course is a continuation of Capstone I.</p> |                   |

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| INFS4339                                                                                                                                                                         | Principles of Web Design |
| Credit Hours                                                                                                                                                                     | 3                        |
| Term Offered                                                                                                                                                                     | Spring                   |
| Prerequisite                                                                                                                                                                     | -                        |
| <p>This course is delivered asynchronously online and is designed with a project-based learning approach. Core content is delivered as exercises and videos leading to small</p> |                          |



self-check homework assignments and four larger “show-what-you-know” projects spaced throughout the semester.

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| INFS4340                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Web Programming                                                                                         |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                                                                                       |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Spring                                                                                                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fundamentals of Database Management System (DBMS) (INFS2329) and Object Oriented Programming (COMP1402) |
| <p>This course is about designing and developing dynamic websites using PHP and MySQL. It is more of a hands-on course with in-class practices and take-home assignments. Students will first go through the basics of web development and HTML. Then, the three-tier web architecture for web-based application development will be introduced, discussed, and implemented. Throughout the course, students will learn how to build dynamic web pages and active content by integrating HTML with server-side scripting languages such as PHP and MySQL for the database.</p> |                                                                                                         |

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| INFS4341                                                                                                                                                                                                                                                                                             | Database Design                                              |
| Credit Hours                                                                                                                                                                                                                                                                                         | 3                                                            |
| Term Offered                                                                                                                                                                                                                                                                                         | Spring                                                       |
| Prerequisite                                                                                                                                                                                                                                                                                         | Fundamentals of Database Management System (DBMS) (INFS2329) |
| <p>This is a broad coverage course which gives students a detailed examination of the use of database management systems. The topics covered include conceptual design, data models, query languages, logical database design, physical database design, and introduction to query optimization.</p> |                                                              |



## Course Catalog – Mechanical Engineering Program

Last updated: Spring 2024

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| MECH1101                                                                                                                                                               | MATLAB I |
| Credit Hours                                                                                                                                                           | 1        |
| Term Offered                                                                                                                                                           | Spring   |
| Prerequisite                                                                                                                                                           | -        |
| The course is an introduction to MATLAB programming environment, arrays, creating and running script files, 2D plotting features, functions, and programming elements. |          |

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| MECH2302                                                                                                                                                                                                                                                                                                                                                                                              | Engineering Graphics                        |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                          | 3                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                          | Fall                                        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                          | Calculus & Analytical Geometry I (GMAT1505) |
| Introduction to principles of manual engineering drawing and computer-aided drafting using CAD/SOLIDWORKS software to generate two- and three-dimensional drawings based on the conventions of engineering graphical communications; topics include spatial relationships, multi-view projections and sectioning, dimensioning, graphical presentation of data and fundamentals of computer graphics. |                                             |

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| ENGR2302                                                                                                                                                                                                                                                                       | Statics                                                                                     |
| Credit Hours                                                                                                                                                                                                                                                                   | 3                                                                                           |
| Term Offered                                                                                                                                                                                                                                                                   | Fall                                                                                        |
| Prerequisite                                                                                                                                                                                                                                                                   | Calculus & Analytical Geometry II (GMAT2506) and General Physics with Calculus I (GSCI1405) |
| Basic theory of engineering mechanics using calculus, involving the description of forces, moments, and couples acting on stationary structures; equilibrium in two and three dimensions; free-body diagrams; friction; centroids; centers of gravity; and moments of inertia. |                                                                                             |

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| MECH2107                                                                                                                                                                                                                                                          | MATLAB II           |
| Credit Hours                                                                                                                                                                                                                                                      | 1                   |
| Term Offered                                                                                                                                                                                                                                                      | Fall                |
| Prerequisite                                                                                                                                                                                                                                                      | MATLAB I (MECH1101) |
| The course is a continuing the MATLAB I course. It covers two-dimensional arrays, manipulation of arrays, plots with special graphics, 3D plots, in-line functions, solving a nonlinear equation with one variable, finding the maximum or minimum of a function. |                     |

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| MECH2305                                                                                                                                                                                                                                                                                                                 | Thermodynamics                             |
| Credit Hours                                                                                                                                                                                                                                                                                                             | 3                                          |
| Term Offered                                                                                                                                                                                                                                                                                                             | Spring                                     |
| Prerequisite                                                                                                                                                                                                                                                                                                             | General Physics with Calculus I (GSCI1405) |
| Foundation in thermodynamics and energy-related applications with final aim of performance analysis of thermodynamics engineering devices through the principles of thermodynamics, vapor power and gas power cycles, entropy, and pure substances properties in open, closed system that is reversible or irreversible. |                                            |

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| MECH2308                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Electrical Circuits + Laboratory            |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | General Physics with Calculus II (GSCI2406) |
| This is the first course to electronic circuit. It is required that students have taken General Physics with Calculus II where the students were exposed to basic electronics before enrolling in this class. Students learn electronic elements such as resistor, capacitor, inductor, etc., and circuit analysis of current and voltage dividers, node voltage and mesh current analysis of circuits, Thevenin and Norton equivalents are introduced. More advanced circuit analysis, transient response, and more electronic elements such as diodes, operational amplifiers, transistors, etc., and logic are thought. The materials are also including electromagnetic fields, electric power, transformers, magnetic materials, generators and motors. |                                             |

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| MECH2306                                                                                                                                                                                                                                                                                           | Dynamics           |
| Credit Hours                                                                                                                                                                                                                                                                                       | 3                  |
| Term Offered                                                                                                                                                                                                                                                                                       | Spring             |
| Prerequisite                                                                                                                                                                                                                                                                                       | Statics (ENGR2302) |
| Basic theory of engineering mechanics using calculus, involving the motion of particles, rigid bodies, and systems of particles; Newton's laws; work and energy relationships; principles of impulse and momentum; application of kinetics and kinematics to the solution of engineering problems. |                    |

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| MECH2232                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Manufacturing Processes for ME |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2                              |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Spring                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -                              |
| Introduction to theory of manufacturing techniques and processes with an emphasis on metals, polymers and ceramics working processes for industrial applications. The course discusses various manufacturing processes and forms basic skills in the operation of standard machine tools and manufacturing processes. The instructional parts include introduction to materials engineering, casting, forming, conventional and non-conventional machining and joining. In addition, metrology aspect in manufacturing process is also discussed. |                                |

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| MECH2104                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Machine Shop |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Spring       |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -            |
| Introduction to theory of manufacturing techniques and processes with an emphasis on metals, polymers and ceramics working processes for industrial applications. The course discusses various manufacturing processes and forms basic skills in the operation of standard machine tools and manufacturing processes. The instructional parts include introduction to materials engineering, casting, forming, conventional and non-conventional machining and joining. |              |

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| MECH3313                                                                                                                                                                                                                                                                                                                                                          | Engineering Analysis              |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                      | 3                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                      | Fall                              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                      | Differential Equations (GMAT2308) |
| The course introduces advanced mathematics subjects, covering linear algebra, matrix eigenvalue problems, Fourier series, eigenfunctions, Laplace and Fourier transforms, and applications to ordinary and partial differential equations. The teaching utilizes problem base learning with numerous exercises, so that class participation is highly encouraged. |                                   |

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| MECH3312                                                                                                                                                                                                                                                                          | Introduction to Fluid Mechanics                                                      |
| Credit Hours                                                                                                                                                                                                                                                                      | 3                                                                                    |
| Term Offered                                                                                                                                                                                                                                                                      | Fall                                                                                 |
| Prerequisite                                                                                                                                                                                                                                                                      | Differential Equations (GMAT2308), Thermodynamics (MECH2305) and Dynamics (MECH2306) |
| Fundamentals of fluid mechanics covering properties of fluids, fluid statics, dynamics of incompressible viscous and inviscid flows, control volume formulations of continuity, momentum and energy equations, dimensional analysis, viscous pipe flow, boundary layers and drag. |                                                                                      |

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| MECH3309                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mechanical Behavior of Engineering Materials |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fall                                         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Statics (ENGR2302)                           |
| This course presents the fundamental concepts and quantities necessary to analyze the state of a deformable solid or structure subject to static loading, such as uniaxial loading, bending, and torsion.<br>By the end of the class, the student should have a clear physical understanding of various quantities, such as stresses, strains, failure criteria, safety factor, deflection, buckling, and strain energies, in analyzing general engineering components (e.g., beams, columns). Then, the student should also be able to analyze these quantities, |                                              |

and the relation between them to solve a mechanical design problem and formulate the proper solution.

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| MECH3111                                                                                                                                                                                                                                                                                                                                                                                                     | Mechanics of Materials Laboratory |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                 | Fall                              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                 | -                                 |
| <p>The purpose of this laboratory course is to reinforce the knowledge learned in MECH 3309 (Mechanical Behavior of Engineering Materials) and MECH 3310 (Fundamentals of Materials for Engineers) by providing hands-on experience with the type of equipment commonly used in labs in engineering development and research, as well as the types of tests used to determine the strength of materials.</p> |                                   |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MECH3310                                                                                                                                                                                                                                                      | Fundamental of Materials for Engineers                                                                                                                                |
| Credit Hours                                                                                                                                                                                                                                                  | 3                                                                                                                                                                     |
| Term Offered                                                                                                                                                                                                                                                  | Fall                                                                                                                                                                  |
| Prerequisite                                                                                                                                                                                                                                                  | General Chemistry I (GSCI1303) + General Chemistry I Lab (GSCI1103L) and General Physics with Calculus I (GSCI1405) + General Physics with Calculus I Lab (GSCI1105L) |
| <p>This course explains the principles which underlie and relate the behavior, properties and processing of materials to their engineering applications. The instructional parts include structure, mechanical properties and various class of materials.</p> |                                                                                                                                                                       |

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| MECH3317                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Dynamics of Machines |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Spring               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Dynamics (MECH2306)  |
| <p>This course presents the fundamental concepts and quantities necessary to perform kinematic design and dynamic analysis of mechanisms (linkages) and machines. In the class, the following items will be covered:</p> <ul style="list-style-type: none"> <li>● Various classes of linkages, concepts of mobility and function,</li> <li>● Basic graphical and analytical methods of linkage synthesis,</li> <li>● Linkage design and analysis software package,</li> <li>● Position, velocity, acceleration and force analysis on linkages and machines,</li> <li>● Basic balancing techniques of rotating machinery</li> </ul> |                      |

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| MECH3233                                                                                                                                                                                                                                                          | Mechatronics & Laboratory                   |
| Credit Hours                                                                                                                                                                                                                                                      | 2                                           |
| Term Offered                                                                                                                                                                                                                                                      | Spring                                      |
| Prerequisite                                                                                                                                                                                                                                                      | Electrical Circuits + Laboratory (MECH2308) |
| <p>The course is an advanced class of the Electrical Circuit and emphasizes more on practical laboratory activities so that students have direct hands-on experience on theory and application. It is required that students have taken the course MECH 2308,</p> |                                             |



Electrical Circuits + Laboratory, where the students were exposed to basic electronics before enrolling this class. The students will learn microcontroller and programming, sensors, DC and stepper motors, Pulse Width Modulation (PWM), etc.

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| MECH3316                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Instrumentation Laboratory                                                |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                                                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Spring                                                                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Thermodynamics (MECH2305) and Electrical Circuits + Laboratory (MECH2308) |
| In this course, students will learn about various things related to measurement systems: behavior of instruments, how instruments work, statistics, error propagation, and others. Any automatic control (feedback control) system needs a measurement system and the performance of the feedback control system is affected by the performance of the measurement instruments. By learning this course, students will have an idea of the importance of understanding the principle of measurement. Theoretical discussion and laboratory activities will be done in this course. |                                                                           |

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| MECH3315                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Engineering Component Design                                                                                         |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                                                                                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Spring                                                                                                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Engineering Graphics (MECH2302), Machine Shop (MECH2104) and Mechanical Behavior of Engineering Materials (MECH3309) |
| This course presents the application of failure analysis methods to the design of specific machine components such as shaft, joints/bolted/riveted/welded joints, gear, spring and slender/thin-walled structures. After attending the class, students will (1) understand the main steps of the mechanical design process, (2) be familiar with fundamental notions of fatigue and static failure, and (3) able to apply these notions to the design of specific components such as shafts and joints. |                                                                                                                      |

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| MECH3234                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mechanical Design Project                                                                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2                                                                                                          |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spring                                                                                                     |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mechanical Behavior of Engineering Materials (MECH3309), Thermodynamics (MECH2305) and Dynamics (MECH2306) |
| In this course, students will design and analyze mechanical engineering machines starting with identifying needs, defining problems, designing, planning manufacturing methods and presenting design prototypes. Mechanical engineering design depends on the field of research owned by mechanical engineering lecturers and students can choose the field they want. In engineering design, students are expected to be able to produce prototype designs that are equipped with electrical and control systems. After attending lectures, students will (1) understand the main steps of engineering design, (2) be able to classify preparations prior to the manufacturing process, and (3) |                                                                                                            |

develop prototype designs and be able to explain the manufacturing process of prototypes.

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| MECH3318                                                                                                                                                  | Numerical Methods                                          |
| Credit Hours                                                                                                                                              | 3                                                          |
| Term Offered                                                                                                                                              | Spring                                                     |
| Prerequisite                                                                                                                                              | MATLAB II (MECH2107) and Differential Equations (GMAT2308) |
| Introduction to linear algebra; solution of engineering problems based upon an integrated approach combining numerical analysis and the use of computers. |                                                            |

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| STEM4301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Internship             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Min. 100 credits taken |
| Internships integrate knowledge and theoretical concepts learned from the academic setting with practice application and personal and career skill development in a professional setting. Students must first search for a company/ corporation/ establishment where they will conduct the internship. The business nature of the company/ corporation/ establishment must be related to mechanical engineering. During the internship, students will perform the tasks given by their supervisor in the company/corporation/ establishment; and students will interact with other employees. The interaction between the students and the related people in the company/ corporation/ establishment where they perform their internship will form the personal and career skill development in a professional setting; while performing the tasks given by their supervisor will make the students apply the knowledge they gained in the academic setting. |                        |

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| STEM4302                                                                                                                                                                                                                                                                                     | Senior Capstone I |
| Credit Hours                                                                                                                                                                                                                                                                                 | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                 | Fall              |
| Prerequisite                                                                                                                                                                                                                                                                                 | -                 |
| The projection of the planned designed into prototype, design and working system testing and iterative evaluation prior prototype finalization. Assessment will be conducted based on the progress and the deliverance of the program outcomes through oral presentation and written report. |                   |

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| MECH4125                                                                                                                                                                                                                                              | Senior Colloquium |
| Credit Hours                                                                                                                                                                                                                                          | 1                 |
| Term Offered                                                                                                                                                                                                                                          | Fall              |
| Prerequisite                                                                                                                                                                                                                                          | -                 |
| This course provides a transition between the academic experience and the professional world. Lectures on several issues, such as interviewing, resume writing, becoming a registered PE, business consulting, engineering ethics, as well as updated |                   |

topics such as digital industry, Industry Revolution 4.0, Technopreneurship, and manufacturing world are provided. Professionals in each field are invited to share their experiences. In the class, the learning process will be done by active learning approach, such as by conducting focus group discussion based on the materials given by speakers, problem solving of real case, reading book assignment, and making an essay about the student's vision for their professional period.

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| MECH4321                                                                                                                                                                                                                                                                                                     | Control System Design                                   |
| Credit Hours                                                                                                                                                                                                                                                                                                 | 3                                                       |
| Term Offered                                                                                                                                                                                                                                                                                                 | Fall                                                    |
| Prerequisite                                                                                                                                                                                                                                                                                                 | Engineering Analysis (MECH3313) and Dynamics (MECH2306) |
| Mathematical modeling of dynamical systems, hardware and software issues; computer simulations; classical control methods including transient response, steady-state errors, bode diagrams, root locus and design of closed loop control systems; introduction to state feedback design and digital control. |                                                         |

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| MECH4226                                                                                                                                | Senior Mechanical Laboratory          |
| Credit Hours                                                                                                                            | 2                                     |
| Term Offered                                                                                                                            | Fall                                  |
| Prerequisite                                                                                                                            | Instrumentation Laboratory (MECH3316) |
| Hands-on experience involving application of fundamental principles in intermediate thermodynamics, fluid mechanics, and heat transfer. |                                       |

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| MECH4319                                                                                                  | Heat Transfer                                                                                          |
| Credit Hours                                                                                              | 3                                                                                                      |
| Term Offered                                                                                              | Fall                                                                                                   |
| Prerequisite                                                                                              | Thermodynamics (MECH2305), Introduction to Fluid Mechanics (MECH3312) and Numerical Methods (MECH3318) |
| Studies of conduction, convection, and radiation heat transfer with applications to engineering problems. |                                                                                                        |

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| MECH4320                                                                                                                                   | Intermediate Thermodynamics |
| Credit Hours                                                                                                                               | 3                           |
| Term Offered                                                                                                                               | Fall                        |
| Prerequisite                                                                                                                               | Thermodynamics (MECH2305)   |
| Intermediate level of thermodynamics course that involves power systems, non-reacting and reacting mixtures, psychrometrics, gas dynamics. |                             |



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| STEM4303                                                                                                                                                                                                                                                                                                                                                               | Senior Capstone II |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                           | 3                  |
| Term Offered                                                                                                                                                                                                                                                                                                                                                           | Spring             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                           | Senior Capstone I  |
| <p>This course is a continuation of the Senior Capstone I (STEM4302). The projection of the planned designed into prototype, design and working system testing and iterative evaluation prior prototype finalization. Assessment will be conducted based on the progress and the deliverance of the program outcomes through oral presentation and written report.</p> |                    |

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| MECH4324                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mechanical Vibrations |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Spring                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -                     |
| <p>This course will let the students explore and apply mathematical concepts and scientific principles to identify, solve, and analyze problems dealing with vibration systems. Emphasis will be placed on both free and forced vibrations of simple mechanical systems and the effects of dampening. Multi-degree-of-freedom vibrating systems and the modern methods and tools for finding solutions to a variety of vibration problems will also be discussed. Students will be building essential engineering skills through participation in a variety of learning activities that include video lectures, group work and discussions, conceptual quiz questions, and quantitative homework sets.</p> |                       |

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| MECH4329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Planar Multibody Dynamics with Applications |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Spring                                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Dynamics of Machines (MECH3317)             |
| <p>This course introduces students to computational methods for analyzing simple and complex planar mechanical systems, such as mechanisms and machines. Several analytical methods for formulating the equations of motions are discussed. Students are guided to implement these equations in MATLAB programs to simulate the dynamics of a variety of planar multibody systems. Several well-developed MATLAB programs are also provided to students for modeling and analyzing realistic mechanical systems that otherwise could not be studied with the traditional pencil-and-paper approach.</p> |                                             |



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| MECH4327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Additive Manufacturing                                                                                                                      |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3                                                                                                                                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Spring                                                                                                                                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Engineering Graphics (MECH2302), Machine Shop (MECH2104), Fundamentals of Materials for Engineering (MECH3310) and Heat Transfer (MECH4319) |
| <p>In this course, engineering materials and their properties are first reviewed. Traditional manufacturing such as casting, forming, machining, and joining processes are introduced and discussed next. Additive manufacturing is then presented. Both general process chain and specific processes are presented (e.g. photopolymerization, powder bed fusion process). Materials properties of each manufacturing process are examined and compared to each other. Design and optimization for AM is highlighted. Real engineering applications are reviewed and discussed.</p> |                                                                                                                                             |

## Course Catalog – Visual Communication Design Program

Last updated: Spring 2024

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| VCDD1402                                                                                                                                                                                                                                                                                                                                                              | Narrative Concepts and Storytelling |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                          | 4                                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                          | Fall                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                          | -                                   |
| Students will be introduced to the concepts and outcomes of different narrative forms, both from a historical and theoretical scope. This course focuses on understanding the wider cultural contexts that inform narratives. Students will develop a personal storytelling structure that can be interacted with by others through a range of different assignments. |                                     |

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| GHUM1306                                                                                                                                                                                                                                                                                                                                                                                                      | Arts: Basic Fundamental of Drawings |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                  | -                                   |
| This is an introductory drawing course that is intended to help develop basic observational and perceptual drawing skills. The course teaches students a variety of concepts and approaches in drawing so that the student is able to demonstrate a range of abilities and versatility with technique. Such conceptual variety can be demonstrated through either the use of one or the use of several media. |                                     |

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| VCDD1301                                                                                                                                                                                                                                                                       | The Origin of Design |
| Credit Hours                                                                                                                                                                                                                                                                   | 3                    |
| Term Offered                                                                                                                                                                                                                                                                   | Fall                 |
| Prerequisite                                                                                                                                                                                                                                                                   | -                    |
| This course will encourage students to work independently and in groups to establish an understanding of Design Fundamentals including design's method and process. Comparing design practice in the past and present, interpreting as to reflect design is a way of thinking. |                      |

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| VCDD1229                                                                                                                                                                                                                       | Color & Material |
| Credit Hours                                                                                                                                                                                                                   | 2                |
| Term Offered                                                                                                                                                                                                                   | Spring           |
| Prerequisite                                                                                                                                                                                                                   | -                |
| Identifying, analyzing, exploring, perceiving, philosophizing colors, harmony colors, opaque colors, translucent colors (transparent colors), RGB/CMYK colors, light colors, and color sharing (Primary, Secondary, Tertiary). |                  |

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| VCDD1325                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digital Photography & Videography |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Spring                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -                                 |
| <p>This course introduces students to the core concepts of digital photographic techniques, how-to-use photography software, develop an efficient workflow and archives management. This course explores the aesthetics aspects, basic scientific foundations, and expand the student's knowledge of photography and videography as art forms and as important mediums in commercial utilizations, journalism and news, science, and new media industry.</p> |                                   |

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| VCDD1326                                                                                                                                                                                                                                                                                | Design Thinking & Theory |
| Credit Hours                                                                                                                                                                                                                                                                            | 3                        |
| Term Offered                                                                                                                                                                                                                                                                            | Spring                   |
| Prerequisite                                                                                                                                                                                                                                                                            | -                        |
| <p>This course will give an understanding of the principles of innovation and the process of how to deliver innovation through Design Thinking. Students will practice the five phases of Design Thinking to tackle simple problems that produce solution that leads to innovation.</p> |                          |

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| VCDD1328                                                                                                                                                                                                                                                                                                                                                                                                                      | Typography Fundamental |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                  | Spring                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                  | -                      |
| <p>This course Introduces student to the core elements of typography and understand the poetic potential of type, its spatial and layout-based relationship from the printed page to moving image. Students are introduced to biographies and works of important typographers. The students will develop a personal experience of type, establishes a digital learning journal to identify research and outcomes clearly.</p> |                        |

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| VCDD1427                                                                                                                                                                       | Computer Graphics Design |
| Credit Hours                                                                                                                                                                   | 4                        |
| Term Offered                                                                                                                                                                   | Spring                   |
| Prerequisite                                                                                                                                                                   | -                        |
| <p>Providing an understanding of Computer Graphics Design, a basic course that introduces the software commonly used in practice: Adobe Photoshop &amp; Adobe Illustrator.</p> |                          |

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| VCDD2330                                                                                                                                                                                                                                                    | Fundamental of Audio Video Production |
| Credit Hours                                                                                                                                                                                                                                                | 3                                     |
| Term Offered                                                                                                                                                                                                                                                | Fall                                  |
| Prerequisite                                                                                                                                                                                                                                                | -                                     |
| Provides a basic understanding of producing videos for a variety of uses. Topics include analyzing the pre-production, production, and post-production process, as well as explore the equipment and techniques used to develop a quality video production. |                                       |

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| VCDD2332                                                                                                                                                                                                                                                                                                                                                                                                                             | Visual Communication & Advertising |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                  |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                         | Fall                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                  |
| The course provides the knowledge of working in an advertising agency. While seeking creative concepts that will best convey the advertising message, this course will investigate different subjects which have to do with the professional skills of the Art Designer in an advertising agency e.g. focusing of the advertising promise, optimal photography for conveying the message, typography, layout, use of media and more. |                                    |

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| VCDD2431                                                                                                                                                                                                                                                                                        | 3D for Design |
| Credit Hours                                                                                                                                                                                                                                                                                    | 4             |
| Term Offered                                                                                                                                                                                                                                                                                    | Fall          |
| Prerequisite                                                                                                                                                                                                                                                                                    | -             |
| This course provides the process of taking a shape and molding it into a completed 3D model and animation. The most typical means of creating a 3D model is to take a simple object, called a primitive, and extend or "grow" it into a shape that can be refined and detailed then animate it. |               |

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| VCDD2433                                                                                                                                                                                                                                                                                                                                                                          | Animation & Motion Graphics |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                      | 4                           |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                      | Spring                      |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                      | -                           |
| In this introductory course, students learn to put static (non-moving) images into motion. These courses teach students to use various industry-standard Adobe software programs. Skills developed include manipulation of images, simulation of camera angles and movement, motion contrasting and digital video, this course also involve sound editing and image manipulation. |                             |

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| VCDD2434                                                                                                                                                                                           | VFX & Compositing |
| Credit Hours                                                                                                                                                                                       | 4                 |
| Term Offered                                                                                                                                                                                       | Spring            |
| Prerequisite                                                                                                                                                                                       | -                 |
| This course will explore the latest technological advances in compositing and visual effects that are used in broadcast industries and understanding how 2D and 3D animation integrate with video. |                   |

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| VCDD2335                                                                                                                                                                                                                                                                                                                                           | Media History |
| Credit Hours                                                                                                                                                                                                                                                                                                                                       | 3             |
| Term Offered                                                                                                                                                                                                                                                                                                                                       | Spring        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                       | -             |
| This course focuses on the history of the world mass media - on people, institutions and ideas that describe the past also will influence the future. In this subject there will be treated different topics related to the history of media, including technological changes, social roles, and changes of business structures of the mass media. |               |

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| VCDD4427                                                                                                                                                                                                                                    | Web Design |
| Credit Hours                                                                                                                                                                                                                                | 4          |
| Term Offered                                                                                                                                                                                                                                | Spring     |
| Prerequisite                                                                                                                                                                                                                                | -          |
| This course will provide knowledge on the different areas of web development include web design; interface design; authoring, including standardized code and proprietary software; user experience design; and search engine optimization. |            |

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| VCDD3436                                                                                                                                                                                                                                                                                                                                                                                                                                              | Creative Coding |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                          | Fall            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                          | -               |
| Students will be introduced to the concepts and outcomes of different forms of programming languages through a theoretical and practical scope of contents and activities. Following a wider cultural context and developing a personal storytelling structure, students will join creative practice with aesthetics to affect a more comprehensive understanding of new digital possibilities and thought in the field of Interactive Digital Media. |                 |

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| VCDD3439                                                                                                                                                                                                                        | Design Research Methodology |
| Credit Hours                                                                                                                                                                                                                    | 4                           |
| Term Offered                                                                                                                                                                                                                    | Fall                        |
| Prerequisite                                                                                                                                                                                                                    | -                           |
| Introduction to research processes, including formulation of research problem, research design, sampling and measurement methods, research proposal writing, literature review, data collection, data processing and reporting. |                             |

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| VCDD3437                                                                                                                                                                                                                                                                            | UI/UX Design |
| Credit Hours                                                                                                                                                                                                                                                                        | 4            |
| Term Offered                                                                                                                                                                                                                                                                        | Fall         |
| Prerequisite                                                                                                                                                                                                                                                                        | -            |
| The UI/UX Design course will teach student a design-centric approach to user interface and user experience design, and offers practical, skill-based instruction centered around a visual communications perspective, rather than on one focused on marketing or programming alone. |              |

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| VCDD3438                                                                                                                                                                                                                                                    | IoT in New Media |
| Credit Hours                                                                                                                                                                                                                                                | 4                |
| Term Offered                                                                                                                                                                                                                                                | Fall             |
| Prerequisite                                                                                                                                                                                                                                                | -                |
| This course is designed to teach students about IOT as new media in a design environment. Students will learn about IOT concept and shifting that happens in the design environment. The use of social media and multimedia will be catered in this course. |                  |

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| VCDD3345                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Indonesia Art History |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                     |
| The Indonesian Art History course emphasizes a deep conceptual understanding of Indonesian Art historical concepts. Students will develop the essential skills of visual and contextual analysis. By examining works of Art from Indonesia's Diverse culture and the relationships among these works, students develop an understanding of Indonesian artistic traditions. Students analyze works of art in their contexts, considering issues of patronage, gender, politics, religion, and ethnicity. |                       |



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| VCDD3442                                                                                                                                                                                                                                                                                                                                                                          | Design Entrepreneurship                                                                                                  |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                      | 4                                                                                                                        |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                      | Spring                                                                                                                   |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                      | Creative Coding (VCDD3436), UI/UX Design (VCDD3437), IoT in New Media (VCDD3438), Design Research Methodology (VCDD3439) |
| <p>This course will introduce student to the processes of motivation and innovation and of Entrepreneurship. This course will give an understanding of the principles of entrepreneurship including developing business plan; marketing in the global information age; financing and marketing; innovation and creativity, financial, management; and product identification.</p> |                                                                                                                          |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| VCDD3441                                                                                                                                                                                | Game Design                                                                                                              |
| Credit Hours                                                                                                                                                                            | 4                                                                                                                        |
| Term Offered                                                                                                                                                                            | Spring                                                                                                                   |
| Prerequisite                                                                                                                                                                            | Creative Coding (VCDD3436), UI/UX Design (VCDD3437), IoT in New Media (VCDD3438), Design Research Methodology (VCDD3439) |
| <p>A practical introduction to game design and game design concepts, emphasizing the basic tools of game design: paper and digital prototyping, design iteration, and user testing.</p> |                                                                                                                          |

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| VCDD3340                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Experimental Art & Design |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                         |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Spring                    |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                         |
| <p>In this course the student will work independently and design a project brief that meets with their-own aspirations and passions for personal exploration from selecting a theme, problem analysis, design process, issue, or concern, etc. Students are required to consider narrative concepts and storytelling as a main pillar to the nature of their design project. The student can explore any visual outcomes to convey creative ideas. This course also supports the enhancement of independent learning by exploring how to propose, plan, and manage self-initiated project.</p> |                           |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| VCDD3443                                                                                                                                                      | Augmented & Virtual Reality                                                                                              |
| Credit Hours                                                                                                                                                  | 4                                                                                                                        |
| Term Offered                                                                                                                                                  | Spring                                                                                                                   |
| Prerequisite                                                                                                                                                  | Creative Coding (VCDD3436), UI/UX Design (VCDD3437), IoT in New Media (VCDD3438), Design Research Methodology (VCDD3439) |
| <p>In this course you will learn how to build AR and VR apps with Unity and look at the different ways to produce interactive digital media presentation.</p> |                                                                                                                          |

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| STEM4301                                                                                                                                                                                                                                        | Internship |
| Credit Hours                                                                                                                                                                                                                                    | 3          |
| Term Offered                                                                                                                                                                                                                                    | Summer     |
| Prerequisite                                                                                                                                                                                                                                    | -          |
| A formal apprenticeship, work-based training, temporary employment, or other form of a guided professional experience that a student undertakes externally at a company / organization or under the auspices of a professional or practitioner. |            |

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|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VCDD4423                             | Final Project I (Design Object)                                                                                                                                  |
| Credit Hours                         | 4                                                                                                                                                                |
| Term Offered                         | Fall                                                                                                                                                             |
| Prerequisite                         | Experimental Art & Design (VCDD3340), Game Design (VCDD3441), Design Entrepreneurship (VCDD3442), Augmented & Virtual Reality (VCDD3443). Internship (STEM4301). |
| Development of Final Design Project. |                                                                                                                                                                  |

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| VCDD4422                                                                                                                                                                                                                                                                                                                                                                  | Collaborative Design                                                                                                                                             |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                              | 4                                                                                                                                                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                              | Fall                                                                                                                                                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                              | Experimental Art & Design (VCDD3340), Game Design (VCDD3441), Design Entrepreneurship (VCDD3442), Augmented & Virtual Reality (VCDD3443). Internship (STEM4301). |
| In this course the student will work Collaboratively with real-life project and design a solution that meets with the client's objective. The student can explore any visual outcomes to convey creative ideas. This course also supports the enhancement of team-work learning environment by exploring how to propose, plan, solving the problems, and manage the team. |                                                                                                                                                                  |

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| VCDD4344                                                                                                                                                                                                          | Motion Capture |
| Credit Hours                                                                                                                                                                                                      | 3              |
| Term Offered                                                                                                                                                                                                      | Fall           |
| Prerequisite                                                                                                                                                                                                      | -              |
| Intro to motion capture course. Provides the basics principles of Motion Capture & Performance Capture from T-Pose to ROMs and Realtime capture - this course is essential for those new to the world of 'Mocap'. |                |

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|----------------------------------------|-----------------------------------------------------------------------------|
| VCDD4424                               | Final Project II (Senior Thesis)                                            |
| Credit Hours                           | 4                                                                           |
| Term Offered                           | Spring                                                                      |
| Prerequisite                           | Final Project I (Design Object) (VCDD4423), Collaborative Design (VCDD4422) |
| Final year Project II: Thesis Writing. |                                                                             |



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| VCDD4421                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Professional Practice Portfolio                                              |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 4                                                                            |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Spring                                                                       |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Collaborative Design (VCDD4422), Final Project I (Design Object) (VCDD4423). |
| <p>This course will prepare the students to enter professional life. The students will develop a portfolio that has a strong emphasis on relationship between the creative process of digital design and responsibility to the clients and society in relation to achieve effective interaction strategies and communication objectives. This course will give opportunity to the students develop and practice their critical evaluation skills by evaluating their peer portfolio's quality and comparing these evaluations with the professional assessment given by the Lectures.</p> |                                                                              |



## Course Catalog – Psychology Program

Last updated: Spring 2024

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| PSYC1301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Introduction to Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                          |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                       |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                          |
| <p>This course offers several discussions that become fundamental things from psychology study. The study will deliver the material by introducing basic psychology in general, which aims to help students better understand various psychology courses. This course will cover several topics, including psychology history, biopsychology, sensation, and perception, learning theory, intelligence theory, motivation theory, personality theory, and personal development theory. Additionally, this lecture will look at current concerns and developments in societal and psychological discourse.</p> |                            |

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| PSYC1302                                                                                                                                                                                                                                                                                                                         | Developmental Psychology I |
| Credit Hours                                                                                                                                                                                                                                                                                                                     | 3                          |
| Term Offered                                                                                                                                                                                                                                                                                                                     | Fall                       |
| Prerequisite                                                                                                                                                                                                                                                                                                                     | -                          |
| <p>Survey of research and theory in child development. This course examines age-related change in the social, emotional, cognitive, and linguistic domains from infancy to adolescence. Emphasizes the exploration of empirical literature in psychology, biology, and social science as it relates to developmental issues.</p> |                            |

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| PSYC1303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Career Exploration and Professional Development |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                               |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                                            |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -                                               |
| <p>In this course, Psychology majors explore the many and varied career pathways that they may encounter with a bachelor's degree in the field. This course is designed to allow students to apply psychology-specific content and skills to career goals. Students will be challenged to articulate, apply, and exhibit skills derived from psychology to best prepare them for successful careers in the workforce. This course is an Engaged Learning course. It is designed to guide students through experiential learning and reflection on that experience to prepare students for applying what they've learned beyond the classroom and into their future careers.</p> |                                                 |

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| PSYC1310                                                                                                                                                                                                                                                                                                                                 | Psychodiagnostics I (Observation and Interview) |
| Credit Hours                                                                                                                                                                                                                                                                                                                             | 3                                               |
| Term Offered                                                                                                                                                                                                                                                                                                                             | Spring                                          |
| Prerequisite                                                                                                                                                                                                                                                                                                                             | -                                               |
| This course offers a general introduction to psychodiagnostics. It provides a basic knowledge of the tools for psychological assessment. This course will particularly familiarize students with theoretical knowledge as well as practical details of observation and interview, which are the basic tools in psychological assessment. |                                                 |

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| PSYC1306                                                                                                                                                                                                                                                                                                                              | Developmental Psychology II           |
| Credit Hours                                                                                                                                                                                                                                                                                                                          | 3                                     |
| Term Offered                                                                                                                                                                                                                                                                                                                          | Spring                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                          | Developmental Psychology I (PSYC1302) |
| This course will explore change and continuity in the biological, psychological, and social processes associated with adulthood, with an emphasis on late life. Students will have the opportunity to engage older adults in conversation, and experience first-hand the broad range of individual differences that occur with aging. |                                       |

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| PSYC1304                                                                                                                                                                                                                                                                                                                                                                                                                                 | History and Philosophy of Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                    |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                               |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                    |
| The development of the discipline of psychology, primarily in the last hundred years, has been examined in the context of significant events occurring in society and in other disciplines. Discussion and critical evaluation of psychology as a profession and a science and of the major schools of thought: structuralism, functionalism, associationism, behaviorism, Gestalt psychology, psychoanalysis, and cognitive psychology. |                                      |

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| PSYC2309                                                                                                                                                                                                                                                                                                                                                                                                                                       | Phenomenology and Existentialism |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fall                             |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                |
| This course provides philosophical foundations based on phenomenological philosophy (phenomenology) and existential philosophy (existentialism). Furthermore, it gives philosophical foundations based on the impacts of several theoretical psychological perspectives. Finally, this course also explains the direct effect of phenomenology and existentialism on the phenomenology approach and existentialism in qualitative observation. |                                  |



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| PSYC2311                                                                                                                                                                                                                                                                                                                                                             | Psychodiagnostics II (Intelligence Assessment) |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                         | 3                                              |
| Term Offered                                                                                                                                                                                                                                                                                                                                                         | Fall                                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                         | -                                              |
| <p>This course provides an overview of intelligence assessment relating to adults and children. It provides history, development, goal, and the use of intelligence assessment in broad settings. This course will particularly familiarize students with theoretical knowledge as well as practical details of intelligence assessment for adults and children.</p> |                                                |

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| PSYC2308                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Research Methods I (Quantitative) |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3                                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Fall                              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | -                                 |
| <p>Psychology major will gain experience in a range of psychological research methods. This course is intended for students who have completed the Statistics course. In this course, students are guided to understand the quantitative research process, starting from problem formulation, research variable formulation, literature research, data collection techniques, sampling procedures, interpreting research results, preparing proposals and research reports.</p> |                                   |

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| PSYC2326                                                                                                                                                                                      | Industrial and Organization Psychology |
| Credit Hours                                                                                                                                                                                  | 3                                      |
| Term Offered                                                                                                                                                                                  | Spring                                 |
| Prerequisite                                                                                                                                                                                  | -                                      |
| <p>This course offers the application of the principles of psychology to industrial and social organizations, including personnel, human factors, organizational and consumer psychology.</p> |                                        |

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| PSYC2312                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Research Method II (Qualitative) |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Spring                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -                                |
| <p>This course provides a generic introduction to the knowledge, skills, and tools that a student will require when undertaking a qualitative research study in education. In particular, the subject focuses on the areas of data collection and analysis. Emphasis is given to understanding, evaluating, and gaining experience in the multiple processes of gathering, managing, exploring, interrogating, and interpreting textual data. This will include practical sessions in using and evaluating Qualitative Data Analysis Software. Students will be encouraged to apply their skills to data that is relevant to their specialist needs.</p> |                                  |

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| PSYC2313                                                                                                                                                                                                                                                                             | Introduction to Psychology Test |
| Credit Hours                                                                                                                                                                                                                                                                         | 3                               |
| Term Offered                                                                                                                                                                                                                                                                         | Spring                          |
| Prerequisite                                                                                                                                                                                                                                                                         | -                               |
| This course introduces the meaning, process, and methods of psychodiagnostics, as well as the benefits in psychological examinations. This will include general orientations of psychological diagnostics, principles, concepts of using tests and the limitations of them as tools. |                                 |

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| PSYC2221                                                                                                                                                                                                                                                                                                                                                                                                     | Code of Ethics |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                 | 2              |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                 | Spring         |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                 | -              |
| This course provides students with an overview of ethical issues that arise in psychological practice, teaching, and research. Students will be familiarized with the codes of ethics (e.g. Indonesian Psychological Association (HIMPSI)'s Code of Ethics and American Psychological Association (APA)'s Code of Ethics) and ethical guidelines. It applies both to psychology academics and psychologists. |                |

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| PSYC2333                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Child and Adolescence Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                                |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                              | Spring                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                              | -                                |
| Child and Adolescence Psychology is a subject that has goals examining complex processes of growth and development during childhood and adolescence. The materials include developmental milestone, attachment theory, cognitive development, social development, emotional development, psychopathology, parenting and family dynamics, the influences of culture on parenting and development and assessment and intervention for child and adolescence |                                  |

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| PSYC2334                                                                                                                                                                                                                                                                                                     | Promoting Healthy Aging |
| Credit Hours                                                                                                                                                                                                                                                                                                 | 3                       |
| Term Offered                                                                                                                                                                                                                                                                                                 | Spring                  |
| Prerequisite                                                                                                                                                                                                                                                                                                 | -                       |
| Promoting Healthy Aging is a class that will provide students with knowledge and skills for helping students understand how to help growing older individuals to enjoy their late adulthood years, with physical, emotional, and above all psychological health, with all social and cultural repercussions. |                         |



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| PSYC3323                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Cognitive Neuroscience |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Fall                   |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -                      |
| <p>This course is, in essence, a user guide to your brain. How do you store a lifetime of memories, make difficult decisions, and understand and generate language? What factors drive how you perceive the world and engage in a wide range of motor actions from taking a single step to dancing? What enables you to feel emotions and understand what others are thinking? The main objective of this course is for you to advance your knowledge of how the brain supports these cognitive abilities and others. Major topics will include sensation and perception, motor control, attention, learning and memory, language, executive functions, and social cognition. To cover these topics, you will be exposed to multiple methods for studying the brain, including a variety of modern brain imaging techniques as well as examination of individuals with brain lesions.</p> <p>Cognitive neuroscience is one of the most fascinating and active fields of research today. By the end of this course, we hope that you will have a better understanding of this rapidly advancing field, what we presently know, and what big questions are still under scientific investigation.</p> |                        |

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| PSYC3315                                                                                                                                                                                                                                                                                                                       | Clinical Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                   | 3                   |
| Term Offered                                                                                                                                                                                                                                                                                                                   | Fall                |
| Prerequisite                                                                                                                                                                                                                                                                                                                   | -                   |
| <p>Clinical Psychology teaches comprehensive mental and behavioral health care for individuals, couples, families, and groups, and how human beings can maintain themselves without any mental health issues. Clinical psychology also concerns with the assessment and treatment of mental illness and abnormal behavior.</p> |                     |

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| PSYC4327                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Health Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3                 |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fall              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -                 |
| <p>The main aim of Health Psychology is to introduce students to the broad theories, terminology, and substantive research findings of the field through consideration of the psychological influences on illness and wellness, as well as the effects of physical health on psychosocial functioning. Throughout the term, we will examine how these processes operate across varied levels of analysis, including, for example, consideration of the different biological systems of the body, how people make health behavior changes, the effects of interpersonal contexts on physical health, the development, maintenance, and psychosocial correlates of specific diseases, cultural variation in disease and health behaviors, and the role of larger systems in promoting wellness and the management of chronic illness.</p> |                   |



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| PSYC3320                                                                                                                                                                                                                                              | Social Psychology |
| Credit Hours                                                                                                                                                                                                                                          | 3                 |
| Term Offered                                                                                                                                                                                                                                          | Fall              |
| Prerequisite                                                                                                                                                                                                                                          | -                 |
| This course as an introduction to the major theories and research findings of social psychology. Specific topics covered in the class include self, social cognition, attitudes, interpersonal relations, group processes, prejudice, and aggression. |                   |

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| PSYC3318                                                                                                                                                                                                                                                                                                                                        | Psychodiagnostics III (Personality Assessment) |
| Credit Hours                                                                                                                                                                                                                                                                                                                                    | 3                                              |
| Term Offered                                                                                                                                                                                                                                                                                                                                    | Fall                                           |
| Prerequisite                                                                                                                                                                                                                                                                                                                                    | -                                              |
| This course is the third level of psychodiagnostics, which offers an overview of personality assessment. It provides history, development, goal, and the use of personality assessment in broad settings. This course will particularly familiarize students with theoretical knowledge as well as practical details of personality assessment. |                                                |

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| PSYC4329                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Abnormal Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                   |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Spring              |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                   |
| Abnormal psychology is a branch of psychology that empirically studies abnormalities in cognition, emotion, and behavior. This course is intended to provide a broad overview of mental illness, with a special emphasis on the classification of mental illness by the Diagnostic and Statistical Manual 5 (DSM-5). The course will also cover theoretical approaches to treating mental health problems and specific treatments that work and do not work for different diagnoses. In addition to learning about diagnosis and treatment, students will be introduced to the biopsychosocial model and the causal theories behind. |                     |

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| PSYC3317                                                                                                                                                                                                                                                                                                                                                                                                      | Educational Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                  | 3                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                  | Spring                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                  | -                      |
| Educational psychology is designed to give the student a thorough understanding of the theories and principles of psychology as applied to teaching and learning. The students will demonstrate knowledge, understanding, and application of theories and principles of development, learning, memory, motivation, individual differences, instruction, classroom management, and measurement and evaluation. |                        |

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| PSYC3324                                                                                                                                                                     | Forensic Psychology |
| Credit Hours                                                                                                                                                                 | 3                   |
| Term Offered                                                                                                                                                                 | Spring              |
| Prerequisite                                                                                                                                                                 | -                   |
| Forensic psychology learns the application of psychological knowledge to help answer legal question, raising in criminal, civil, contractual, or other judicial proceedings. |                     |

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| PSYC3307                                                                                                                                                                                                                                                                                                                                                                                                                                             | Psychological Measurement and Statistics               |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3                                                      |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                         | Spring                                                 |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                         | Statistics (GMAT1310) and Social Psychology (PSYC3320) |
| Does sleep improve memory? Does having friends improve mental health? These questions can be examined through controlled scientific experiments. This course will teach students the methods of statistics that are used for exploring data collected in experiments and for evaluating scientific hypotheses. Students will learn how to apply the core statistical tools used in science such as t-tests, ANOVA, regression, and Chi-square tests. |                                                        |

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| PSYC3316                                                                                                                                                                                                                                                                                | Personality Psychology |
| Credit Hours                                                                                                                                                                                                                                                                            | 3                      |
| Term Offered                                                                                                                                                                                                                                                                            | Spring                 |
| Prerequisite                                                                                                                                                                                                                                                                            | -                      |
| This course provided basic concepts and issues in personality theory and research, approaches to personality description and assessment. This course facilitates students to examine the development and experience of an individual based on the perspectives of the various theories. |                        |

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| PSYC3325                                                                                                                                                                                                                                                                                                                                                                                                                      | Biopsychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                  | 3             |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                  | Spring        |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                  | -             |
| This course will give you an in-depth view of the field of clinical psychology, including assessment/diagnosis, interviewing, testing and the different therapies and research on them. It will also cover specialties (child clinical, health, clinical neuropsychology, and forensic psychology) and differentiate clinical from the other professions (social work, psychiatry, marriage and family therapy, counselling). |               |

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| PSYC3326                                                                                                                                                            | Psychology of Entrepreneurship |
| Credit Hours                                                                                                                                                        | 3                              |
| Term Offered                                                                                                                                                        | Spring                         |
| Prerequisite                                                                                                                                                        | -                              |
| The subject learns about the application of psychological concept of helping entrepreneurship organization to achieve success and sustainability in business world. |                                |

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|--------------|-------------------|
| PSYC4331     | Senior Capstone I |
| Credit Hours | 3                 |
| Term Offered | Fall              |
| Prerequisite | -                 |
| Tbd.         |                   |

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| PSYC3322                                                                                                                                                                                                                                                                                                                              | Counseling Psychology |
| Credit Hours                                                                                                                                                                                                                                                                                                                          | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                          | Fall                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                          | -                     |
| This course is an introductory class to Counseling Psychology. Students will learn about the history of the development of counseling, effective counseling processes, and counseling theories. Students will also learn to recognize the characteristics of a counselor who has integrity and upholds a professional code of ethics. |                       |

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| PSYC4328                                                                                                                                                                                                                                             | Mental Health and Mental Wellness |
| Credit Hours                                                                                                                                                                                                                                         | 3                                 |
| Term Offered                                                                                                                                                                                                                                         | Fall                              |
| Prerequisite                                                                                                                                                                                                                                         | -                                 |
| Mental health and mental wellness learn about how a human being thinks, feels and act and how they can handle stress, relate to others and make healthy choices. Mental health and mental wellness is important concepts at every stage of lifespan. |                                   |

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| PSYC4330                                                                                                                                                                                                                                                                                                                                                                                                                                      | Community Development |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                                                                                                  | Fall                  |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                                                                                                  | -                     |
| This thematic internship program is a volunteer and humanitarian activity to help individuals develop awareness of social values in social life. Students are given the opportunity to be involved in various humanitarian activities, such as contributing in the fields of media and information literacy, education, the environment, tourism and the creative economy, as well as other social fields related to the field of psychology. |                       |

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| PSYC3314                                                                                                                                                                                         | Cognitive Development |
| Credit Hours                                                                                                                                                                                     | 3                     |
| Term Offered                                                                                                                                                                                     | Fall                  |
| Prerequisite                                                                                                                                                                                     | -                     |
| Studies the human mind and learns about the processes of attention, perception, sleep, dream, memory, and storage, and higher order thinking that enables human beings to learn about the world. |                       |

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| PSYC4632                                                                                                                                                                                                                                                                                                                                                         | Senior Capstone II (Skripsi & Jurnal) |
| Credit Hours                                                                                                                                                                                                                                                                                                                                                     | 6                                     |
| Term Offered                                                                                                                                                                                                                                                                                                                                                     | Spring                                |
| Prerequisite                                                                                                                                                                                                                                                                                                                                                     | Senior Capstone I (PSYC4331)          |
| A culminating experience for majors involving a substantive project that demonstrates a synthesis of learning accumulated in the major, including broadly comprehensive knowledge of the discipline and its methodologies. Senior standing required. May be repeated: for a total of 6 units of credit. Senior capstone 1 is required to take senior capstone 2. |                                       |

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| PSYC4305                                                                                                                                                                                                                                                                                                                        | Internship |
| Credit Hours                                                                                                                                                                                                                                                                                                                    | 3          |
| Term Offered                                                                                                                                                                                                                                                                                                                    | Spring     |
| Prerequisite                                                                                                                                                                                                                                                                                                                    | -          |
| This course is an internship program, where students will have the opportunity to apply the concepts, theories, and skills that they acquired during their studies. This internship program can be implemented in companies, agencies, and institutions both public and private, especially in positions related to psychology. |            |

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| PSYC3319                                                                                                                                                                                                                                                                                              | Psychology of Leadership |
| Credit Hours                                                                                                                                                                                                                                                                                          | 3                        |
| Term Offered                                                                                                                                                                                                                                                                                          | Spring                   |
| Prerequisite                                                                                                                                                                                                                                                                                          | -                        |
| Subject that studies human behavior and the environment that can help them define what an effective leader. Psychology of leadership will motivate students for developing an ideal type in her or himself to build leader's character, in a good fit with 21 <sup>st</sup> century leaders typology. |                          |